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ARTICLE

Review of cognitive diagnostic models (CDMs): Recent methodological advancements for addressing practical challenges

Chun Wang[1] | **Yale Quan**[2] | **David Arthur**[3]

[1]College of Education, University of Washington, Seattle, Washington, USA

[2]College of Education and Human Sciences, University of Southern Mississippi, Mississippi, USA

[3]Department of Statistics, University of Washington, Tacoma, Washington, USA

Correspondence
Chun Wang, College of Education, University of Washington, 312E Miller Hall, 2012 Skagit Ln, Seattle, WA 98105, USA.
Email: wang4066@uw.edu

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Abstract

Cognitive diagnostic models (CDMs) have become essential tools for providing fine-grained information about individuals' mastery of cognitive skills. While prior reviews have emphasized statistical foundations and deep learning-based developments, this article focuses on recent methodological innovations designed to address persistent challenges in applying CDMs across various contexts. We highlight six areas of advancement: (1) extending models to accommodate polytomous attributes, thereby capturing multiple levels of proficiency; (2) discovering attribute relationships to reveal learning trajectories and support instructional design; (3) accelerating estimation through computational strategies that enable scalability to high-dimensional attributes; (4) improving estimation in small-sample contexts, particularly relevant for classroom applications; (5) refining and estimating Q-matrices through both statistical and machine learning approaches; and (6) evaluating model and item fit. Together, these methodological developments enhance the interpretability, efficiency, and applicability of CDMs, broadening their utility in various types of assessment, larger or smaller scales. The review complements existing surveys by emphasizing innovations that directly address application barriers, and it points to promising future directions including integration with deep learning and large language models.

KEYWORDS

cognitive diagnosis model/diagnostic classification model, item response theory

1 | INTRODUCTION

In education, assessments constructed under a formative paradigm and grounded in research on learner cognition would much better serve the needs of various stakeholders (e.g. teachers, students and parents). To support these next-generation assessments that provide fine-grained feedback for students and teachers (Bradshaw & Templin, 2014; Leighton & Gierl, 2007; Templin & Bradshaw, 2013), cognitive diagnostic models (CDMs) have arisen as advanced psychometric models in the past few decades. They are a family of psychometric models developed to provide detailed information about individuals' mastery of specific cognitive skills or attributes. These models were conceived in response to limitations of traditional psychometric approaches, such as Classical Test Theory (CTT) and Item Response Theory (IRT), which primarily estimate a unidimensional or low-dimensional general ability but offer limited insight into the specific subskills or knowledge components underlying test performance (Embretson, 1985; Rupp et al., 2010). Although the fast emergence of CDMs occurred around the 1990s, the theoretical motivation for these models originates from the convergence of cognitive psychology, psychometrics and educational measurement. Beginning in the 1970s, researchers in cognitive psychology emphasized the importance of assessments that could reflect the structure and processes of human cognition (Glaser, 1994; Mislevy, 1994). Parallel efforts in educational assessment called for diagnostic tools that could support formative feedback and individualized instruction by pinpointing which specific skills students had or had not mastered (Leighton & Gierl, 2007).

This led to the development of *Restricted Latent Class Models (RLCMs)* that form the statistical foundation of CDMs. Unlike general latent class models, which treat the latent structure as unconstrained, RLCMs impose restrictions (often based on a Q-matrix) that encode, a priori, assumptions about the relationships between latent attributes and item responses (DiBello et al., 1995; Haertel, 1989). These constraints allow for interpretable inferences about an individual's mastery profile across a predefined set of binary or categorical skills. During the 1990s and early 2000s, specific CDMs were introduced, most notably the Deterministic Inputs, Noisy ‘And’ gate (DINA) model and its disjunctive counterpart, the Deterministic Inputs, Noisy ‘Or’ gate (DINO) model (Junker & Sijtsma, 2001; Templin & Henson, 2006). These models were further generalized into frameworks such as the Log-Linear Cognitive Diagnosis Model (LCDM; Von Davier, 2014) and the Generalized DINA (G-DINA) model, allowing more flexible representations of item-attribute relationships (De La Torre, 2009). The broader term *Diagnostic Classification Models (DCMs)* was later adopted to encompass the full range of models—including RLCMs—that aim to classify individuals based on discrete latent traits (Rupp et al., 2010). While CDMs are often used interchangeably with DCMs, some distinguish between them by reserving ‘CDM’ for models with a stronger foundation in cognitive theory and ‘DCM’ for the broader class that includes empirically driven or atheoretical models. In this review, we do not distinguish between them.

Methodological research on CDMs has flourished in the past several decades. In parallel, application research that connects the theory of CDMs with practice has emerged in different substantive areas. For instance, they are applied in education over a wide range of developmental phases, including proportional reasoning (Tjoe & Torre, 2012), rational numbers (Bradshaw et al., 2014) and undergraduate Physics (Morphew et al., 2018). Beyond education, CDMs have also been applied to many other domains, such as assessing pathological gamblers (Templin & Henson, 2006), internet addiction (Tu et al., 2017), quality of life in breast cancer patients (Liang et al., 2023) or assessing detailed skill qualifications of job candidates to facilitate a better match between job seekers and job requirements in online recruitment (Qin et al., 2020; Zhu et al., 2018).

One of the earliest reviews of CDM is by Rupp and Templin (2008), and since then, Sessoms and Henson (2018) as well as Ravand and Baghaei (2020) have provided more recent reviews. There are two latest comprehensive reviews of CDMs, one by Zhang et al. (2023) and the other by Wang et al. (2024). Zhang et al. (2023) can be viewed as an extension of Ravand and Baghaei (2020) to focus on the latest (up to 2023) statistical applications of cognitive diagnostic testing, and they review different frameworks for cognitive diagnosis, different probabilistic CDMs and nonparametric methods. Because their publication is in the journal of *Annual Review of Statistics and Its Application*, they devote an entire section to statistical inference, including parameter estimation and model identifiability. In contrast, although Wang et al. (2024) also present a brief review of CDM history by listing a whole host of representative CDMs from the 1960s to the present (i.e. see their figure 4), a large portion of their review is devoted to machine learning-based CDM. It is a broad family of models that were proposed in the recent decade, including both non-deep learning and deep learning-based models. To avoid too much overlap with these two reviews, our paper will omit what are detailed in their papers but instead elaborate and highlight the recent methodological advancements of CDMs, which intend to address the challenges emerged from CDM applications. Note that Sessoms and Henson (2018) also provided a rather recent review and critical commentary on the applications of CDMs, but their focus is on practical challenges in applications such as reliability and validity evidence of CDMs for informing practical decision-making, whereas we focus on three aspects of application needs that are germane to CDM methodology: *modelling*, *estimation* and *validation*. For modelling, we discuss two topics: (1) allowing gradation of attributes and (2) discovering attribute relationships. Regarding estimation, we introduce methods that will (3) speed up computation with high-dimensional attributes and (4) provide accurate model estimation with small sample sizes. Lastly, we introduce validation methods that include (5) Q-matrix discovery and refinement and (6) model fit evaluation.

This article provides a conceptual review of the above six topics. The purpose of the review is not to provide an exhaustive survey of the literature, but rather to synthesize and organize representative modelling and methodological approaches that have shaped the development of the field. Accordingly, the studies discussed were selected based on their relevance to the methodological themes addressed in this paper.[1] Given the rapid growth of CDM literature, it is possible that many additional studies exist that are not explicitly discussed here.

2 | REVIEW OF METHODOLOGICAL ADVANCEMENTS

This section reviews the six topics listed above. Below is a list of basic notations that will be used throughout the review:

l: profile;

L: total number of profiles;

k: attribute;

K: total number of attributes;

M: number of categories (or levels) per attribute [For notational simplicity, throughout the paper, we do not assume varying levels per attribute, although all polytomous CDMs reviewed herein can allow each attribute to have distinctive levels, i.e. M_k];

S: number of response options per item;

j: item;

J: total number of items.

2.1 | Modelling polytomous attributes

Traditional CDMs assume that an individual's latent profile is a discrete constellation of binary attributes that denote the mastery or non-mastery of the fine-grained skills, such as mastering the chain

[1]The goal of the present paper is to provide a conceptual overview of CDMs rather than a systematic or scoping review of the literature. Consequently, we need to acknowledge that the literature included in the review was not identified through a formal database search with predefined inclusion and exclusion criteria. Instead, we used key words such as 'cognitive diagnostic models', 'diagnostic classification models', 'estimation', 'Q-matrix validation' and 'item fit' to identify relevant papers published since 2020 onwards from the major psychometric journals such as *Psychometrika, Journal of Educational and Behavioral Statistics, British Journal of Mathematical and Statistical Psychology, Multivariate Behavioral Research, Journal of Educational Measurement*, as well as a few other journal outlets. As mentioned, we only include methods that are relevant to the themes and that are not extensively discussed in other recent CDM review papers.

rule in calculus, although recently, several studies have also extended the concept of attribute to include 'misconceptions' and thereby, having a misconception means an individual possesses this intermediate understanding (Bradshaw et al., 2014; DiBello et al., 2015; Kuo et al., 2018; Wang, 2024). Still, these models assume attributes to be binary. In contrast, polytomous attributes capture multiple levels of proficiency within an attribute rather than a simple binary classification. This finer granularity enables more precise diagnostic feedback, which can be particularly useful in educational assessment where learning progresses in stages. One concrete example of a polytomous attribute is a three-level attribute related to fractions, with level 0 indicating 'not able to compare and order fractions', level 1 indicating 'able to compare two fractions' and level 2 indicating 'able to order three or more fractions' (Chen & De La Torre, 2013; Ma, 2022).

A number of CDMs for polytomous attributes have been proposed in the literature, including the generalized diagnostic models (GDM, Von Davier, 2008), polytomous attribute RUM (Templin, 2004), polytomous generalized DINA (pG-DINA, Chen & De La Torre, 2013), polytomous diagnostic classification model framework (PDCM, Bao, 2019), higher-order CDM with ordinal attributes (Ma, 2022), saturated polytomous CDM (sp-CDM, De La Torre et al., 2025) and restricted latent class model with polytomous attributes (Wayman et al., 2025), in a chronological order. Rather than reviewing all aspects of these different models, this section highlights *how polytomous attributes are handled by different models*. Other model features, such as condensation rules (e.g. conjunctive, additive and disjunctive) or multivariate relationships among attributes (e.g. higher-order structure), which are not unique to only polytomous models, are omitted here.

Table 1 provides a summary of different ways of handling polytomous attributes, and we provide a more in-depth discussion following Table 1. Unlike CDMs for binary attributes, where $q_{jk} \in \{0, 1\}$, all these models assume that $q_{jk} \in \{0, 1, \dots M\}$, although different models may treat the meaning of "q_{jk}" differently. For instance, in Ma (2022), he provided a mapping matrix (table 1, p. 411) that maps an original q-vector into a reduced one. For example, for an attribute $\boldsymbol{\alpha}$ with four levels (0, 1, 2, 3), the original q-vector for item 1 is (0, 0, 1, 2), meaning that this item cannot distinguish level 0 and level 1 of $\boldsymbol{\alpha}$ but can distinguish the rest. The reduced q-vector for this item becomes $\boldsymbol{q}_1 = (0, 1, 2)$, referring to $\boldsymbol{\alpha} = [0 \; or \; 1], 2, 3$ as reduced three levels. The original q-vector for item 2 is (0, 1, 1, 2), implying that level 1 and 2 of $\boldsymbol{\alpha}$ are not distinguishable by item 2. The reduced q-vector for item 2 becomes $\boldsymbol{q}_1 = (0, 1, 2)$ as well, but it refers to three different levels of $\boldsymbol{\alpha}$ compared to item 1, that is, $\boldsymbol{\alpha} = 0, [1 \; or \; 2], 3$. The flexible models in the last row of Table 1 assume when $q_{jk} = 2$, this item j can distinguish level 0, level 1 and

TABLE 1 Summary of treatment of polytomous attributes in different models.

Models	Polytomous attributes	Remarks
Ordered category attribute coding (OCAC, Karelitz, 2004); pG-DINA (Chen & De La Torre, 2013)	$\alpha^*_{lk} = \begin{cases} 0, \alpha_{lk} < q_{jk} \\ 1, \alpha_{lk} \geq q_{jk} \end{cases}$	When $\alpha_{lk} < q_{jk}$, it does not differentiate non-mastery from partial mastery
GDM (Von Davier, 2008)	$\alpha^*_{lk} = \min\left(q_{jk}, \alpha_{lk}\right)$	A higher level than q_{jk} will not increase the success probability
Higher-order CDM with ordinal attributes (Ma, 2022); sp-G-DINA (De La Torre et al., 2025)	α^*_{lk} is a <u>dummy</u> coded α_{lk}, such as, $\alpha^*_{lk} = \begin{cases} (0,0), \alpha_{lk} = 0 \\ (1,0), \alpha_{lk} = 1 \\ (1,1), \alpha_{lk} = 2 \end{cases}$	This allows for maximum flexibility: (1) mastering different levels of an attribute results in different success probability; and (2) it can model the interactions of different levels of attributes.
Restricted latent class model with order attributes (Wayman et al., 2025)	α^*_{lk} is a <u>cumulative</u> coded α_{lk} with the first column indicating intercept, such as, $\alpha^*_{lk} = \begin{cases} (1,0,0), \alpha_{lk} \geq 0 \\ (1,1,0), \alpha_{lk} \geq 1 \\ (1,1,1), \alpha_{lk} \geq 2 \end{cases}$	

Note: α_{lk} denotes the original polytomous attribute for profile l and attribute k, whereas α^*_{lk} denotes the transformed attributes. α^*_{lk} can take both binary and polytomous form.

level 2 of attribute k, although in Ma (2022), the three levels of attribute k may be associated with different interpretations for different items.

Karelitz (2004) proposed an ordered category attribute coding (OCAC) framework to represent qualitatively ordered mastery levels for each attribute. In this framework, each attribute is coded as an integer starting from 0 and 1 to the highest level, according to its cognitive complexity. Expanding from this framework, Chen and De La Torre (2013) proposed a polytomous generalized DINA (pG-DINA). The primary idea is, for attribute k and a profile l, if $\boldsymbol{\alpha}_{lk} < q_{jk}$ for item j, then $\boldsymbol{\alpha}^*_{lk} = 0$. Here, $\boldsymbol{\alpha}^*_{lk}$ is a collapsed binary attribute and $\boldsymbol{\alpha}^*_{lk} = 1$ only when $\boldsymbol{\alpha}_{lk} \geq q_{jk}$. This treatment simplifies the model such that the classic generalized DINA model with binary attributes can be readily applied on the collapsed attributes. However, this approach assumes that if $q_{jk} = 2$, someone with $\boldsymbol{\alpha}_{lk} = 0$ versus $\boldsymbol{\alpha}_{lk} = 1$ will have the same chance of answering item j correctly, without differentiating non-mastery from partial mastery. In contrast, in Von Davier's (2008) general diagnostic model (GDM), a critical component in the model is $h_k\left(q_{jk}, \boldsymbol{\alpha}_{lk}\right) = \min\left(q_{jk}, \boldsymbol{\alpha}_{lk}\right)$. The rationale is that, since q_{jk} represents a sufficient level for attribute k on item j, a higher level than q_{jk} will not increase the probability of solving item j. When holding other attributes as the same, a skill level lower than q_{jk} results in a lower probability of solving item j. Different from the pG-DINA model, this parameterization distinguishes non-mastery and partial mastery because $\boldsymbol{\alpha}_{lk} = 0$ vs. $\boldsymbol{\alpha}_{lk} = 1$ will lead to different $h_k\left(q_{jk}, \boldsymbol{\alpha}_{lk}\right)$ when $q_{jk} = 2$. In this regard, the pG-DINA model can be viewed as a special case of GDM. One modelling constraint of GDM is that it treats $\boldsymbol{\alpha}_{lk}$ as if it is on the interval scale. That is, when $\boldsymbol{\alpha}_{lk} < q_{jk}$, say when $q_{jk} = 3$, due to the linear term $\lambda_{jk} h_k\left(q_{jk}, \boldsymbol{\alpha}_{lk}\right)$ in the model, the difference in terms of logit of success probability between $\boldsymbol{\alpha}_{lk} = 0$ and $\boldsymbol{\alpha}_{lk} = 1$ is the same as the difference of logit success rate between $\boldsymbol{\alpha}_{lk} = 1$ and $\boldsymbol{\alpha}_{lk} = 2$.

Recently, De La Torre et al. (2025) proposed a generalized CDM for polytomous attributes, and their framework subsumes all extant polytomous CDMs as special cases. In their model, polytomous attributes are dummy coded, so are other recent models (Bao, 2019; Ma, 2022). With such a coding scheme, a saturated model will consider all possible interactions among varying levels. Take item j as an example. Let $K_j = \sum_{k=1}^{K} I\left(q_{jk} > 0\right)$ denote the number of required attributes. The item response function using an *identity* link function is as follows (De La Torre et al., 2025):

$$P_j\left(\boldsymbol{\alpha}_l\right) = \delta_{j0} + \sum_{k=1}^{K_j} \sum_{m=1}^{M-1} \delta_{jkm} I\left[\alpha_{lk} = m\right] + \sum_{k'>k}^{K_j} \sum_{k=1}^{K_j-1} \sum_{m=1}^{M-1} \sum_{m'=1}^{M-1} \delta_{jkmk'm'} I\left[\alpha_{lk} = m\right] I\left[\alpha_{lk'} = m'\right] + \dots$$
$$+ \sum_{m_1=1}^{M-1} \sum_{m_2=1}^{M-1} \cdots \sum_{m_{K_j}=1}^{M-1} \delta_{1m_1 2m_2 \dots K_j m_{K_j}} \prod_{k=1}^{K_j} I\left[\alpha_{lk} = m_k\right] \tag{1}$$

Here in Equation 1, $\boldsymbol{\alpha}_l$ is a reduced vector that only contains relevant attribute for item j, but we omit the subscript j to avoid notational clutter. $\boldsymbol{\delta}_{j0}$ is the intercept and $\boldsymbol{\delta}_{jkm}$ is the main effect of the m th level of the k th attribute. In Equation 1, the terms $I\left[\alpha_{lk} = m\right], \dots, \prod_{k=1}^{K_j} I\left[\alpha_{lk} = m_k\right]$ can be viewed as forming a 'design vector' for δ's, and this design matrix is constructed by the Kronecker product of all $\boldsymbol{d}^*_{lk}$'s, that is, $\boldsymbol{d}^*_{l1} \otimes \boldsymbol{d}^*_{l2} \cdots \otimes \boldsymbol{d}^*_{lK_j}$, where $\boldsymbol{d}^*_{lk} = \left(1, \boldsymbol{\alpha}^*_{lk}\right)$ and $\boldsymbol{\alpha}^*_{lk}$ is a binary coding of the original polytomous $\boldsymbol{\alpha}_{lk}$. Take $K_j = 2$ as an example and assume both $q_{j1} = q_{j2} = 2$, implying that item j measures $\boldsymbol{\alpha}_1$ and $\boldsymbol{\alpha}_2$. Table 2 lists the design vector for different profiles. Note that the last column in Table 2 refers to models based

TABLE 2 Illustration of design matrix in generalized CDM.

$\boldsymbol{d}_{l1}^{*}$	$\boldsymbol{d}_{l2}^{*}$	$\boldsymbol{d}_{l1}^{*} \otimes \boldsymbol{d}_{l2}^{*}$	Implications: Dummy coding	Implications: Cumulative coding
(1, 0, 0)	(1, 0, 0)	(1,0,0,0,0,0,0,0,0)	$\alpha_{l1} = 0, \alpha_{l2} = 0$	$\alpha_{l1} \geq 0, \alpha_{l2} \geq 0$
(1, 0, 0)	(1, 1, 0)	(1,1,0,0,0,0,0,0,0)	$\alpha_{l1} = 0, \alpha_{l2} = 1$	$\alpha_{l1} \geq 0, \alpha_{l2} \geq 1$
(1, 0, 0)	(1, 1, 1)	(1,1,1,0,0,0,0,0,0)	$\alpha_{l1} = 0, \alpha_{l2} = 2$	$\alpha_{l1} \geq 0, \alpha_{l2} \geq 2$
(1, 1, 0)	(1, 0, 0)	(1,0,0,1,0,0,0,0,0)	$\alpha_{l1} = 1, \alpha_{l2} = 0$	$\alpha_{l1} \geq 1, \alpha_{l2} \geq 0$
(1, 1, 0)	(1, 1, 0)	(1,1,0,1,1,0,0,0,0)	$\alpha_{l1} = 1, \alpha_{l2} = 1$	$\alpha_{l1} \geq 1, \alpha_{l2} \geq 1$
(1, 1, 0)	(1, 1, 1)	(1,1,1,1,1,1,0,0,0)	$\alpha_{l1} = 1, \alpha_{l2} = 2$	$\alpha_{l1} \geq 1, \alpha_{l2} \geq 2$
(1, 1, 1)	(1, 0, 0)	(1,0,0,1,0,0,1,0,0)	$\alpha_{l1} = 2, \alpha_{l2} = 0$	$\alpha_{l1} \geq 2, \alpha_{l2} \geq 0$
(1, 1, 1)	(1, 1, 0)	(1,1,0,1,1,0,1,1,0)	$\alpha_{l1} = 2, \alpha_{l2} = 1$	$\alpha_{l1} \geq 2, \alpha_{l2} \geq 1$
(1, 1, 1)	(1, 1, 1)	(1,1,1,1,1,1,1,1,1)	$\alpha_{l1} = 2, \alpha_{l2} = 2$	$\alpha_{l1} \geq 2, \alpha_{l2} \geq 2$

on cumulative coding as in and Wayman et al. (2025). Although the design matrix for dummy coding and cumulative coding is the same, the interpretation is different.

In Equation 1, another point that is worth highlighting is that q-vector is not specifically included therein; instead, the q-vector information is encoded in $\boldsymbol{\alpha}_l$, which represents a reduced vector that only contains attributes measured by the item, regardless of the levels. That is, if this item measures only level 2 of attribute k, and someone with $\boldsymbol{\alpha}_{lk} = 3$ will still have a higher chance of answering the item correctly than others with $\boldsymbol{\alpha}_{lk} = 2$. This monotonicity can be achieved by constraining all δ's be non-negative. De also considered a few constrained versions of the model, such as constraining the maximum success probability to be unchanged when $\boldsymbol{\alpha}_{lk} \geq q_{jk}$.

To illustrate the connection between CDMs with binary and polytomous attributes clearly, let us focus on the additive model. De La Torre et al. (2025) also showed that the fully additive model for polytomous attributes (*f*A-M; Yakar et al., 2021) is a special form of Equation 1 that only retains the intercept and main effects, that is,

$$P_j\left(\boldsymbol{\alpha}_l\right) = \delta_{j0} + \sum\nolimits_{k=1}^{K_j} \sum\nolimits_{m=1}^{M-1} \delta_{jkm} I\left[\alpha_{lk} = m\right]. \tag{2}$$

Because any M-level polytomous attribute can be equivalently represented by $(M-1)$ binary attributes that follow a linear hierarchy, it is easy to demonstrate that Equation 2 is equivalent to the classic additive CDM (a constrained form of G-DINA, De La Torre, 2011) when replacing an M-level polytomous attribute with $(M-1)$ binary attributes. That is,

$$P_j\left(\boldsymbol{\alpha}_l\right) = \delta_{j0} + \sum\nolimits_{k=1}^{K_j} \boldsymbol{\delta}_{jk} \boldsymbol{\alpha}_l^{*} \tag{3}$$

where $\boldsymbol{\alpha}_l^{*}$ is a dummy coded $\boldsymbol{\alpha}_l$, as shown in Table 1. For instance, for an item with a q-vector of (1, 2, 0, 0) and let $\boldsymbol{\alpha}_l$ =(2, 1), then $\boldsymbol{\alpha}_l^{*} = (1, 1, 1, 0)$.

This equivalency between Equations 2 and 3 continues to hold when interaction terms are included, with a caveat that there will be no interactions among dummy-coded attributes within a polytomous attribute. For instance, considering only adding a two-way interaction term to Equation 2, we have the following:

$$P_j\left(\boldsymbol{\alpha}_l\right) = \delta_{j0} + \sum\nolimits_{k=1}^{K_j} \sum\nolimits_{m=1}^{M-1} \delta_{jkm} I\left[\alpha_{lk} = m\right] + \sum\nolimits_{k'>k}^{K_j} \sum\nolimits_{k=1}^{K_j-1} \sum\nolimits_{m=1}^{M-1} \sum\nolimits_{m'=1}^{M-1} \delta_{jkmk'm'} I\left[\alpha_{lk} = m\right] I\left[\alpha_{lk'} = m'\right] \tag{4}$$

Take an example of $K_j = 2$ (assuming item j requires the first two attributes for simplicity) and both $\boldsymbol{\alpha}_{l1}$ and $\boldsymbol{\alpha}_{l2}$ are 3-level polytomous attributes (i.e. $M = 3$). Then from Equation 4, we have

$$P_j\left(\boldsymbol{\alpha}_l = [2,2]\right) = \delta_{j0} + \delta_{j11} + \delta_{j12} + \delta_{j21} + \delta_{j22} + \delta_{j1121} + \delta_{j1221} + \delta_{j1122} + \delta_{j1222} \tag{5}$$

where the last four terms refer to the two-way interaction effects: δ_{j1121} refers to the interaction between $(\boldsymbol{\alpha}_{l1} = 1)$ and $(\boldsymbol{\alpha}_{l2} = 1)$ whereas δ_{j1222} denotes the interaction between $(\boldsymbol{\alpha}_{l1} = 2)$ and $(\boldsymbol{\alpha}_{l2} = 2)$. This exercise aims to reinforce that one can strategically treat each polytomous attribute as a vector of binary attributes with linear constraints, and this is practically useful because one can readily use the estimation algorithms that were originally developed for CDMs with binary attributes directly for polytomous CDMs with minimum modifications. That said, while dummy (or cumulative) coding allows for maximum flexibility, the resulting model is highly parameterized, and Ma (2022) found that adding L1 penalty on δ's would help stabilize model estimation and performance.

2.2 | Discovering attribute relationships

Attribute hierarchy is a core concept in CDM, referring to a directional link among attributes such that some attributes must be mastered before the subsequent attributes can be learned. In educational context, it often reflects theories about students' learning (Dahlgren et al., 2006; Simon & Tzur, 2004), where a piece of knowledge learned earlier serves as a prerequisite for the more advanced knowledge, forming a so-called learning trajectory (Corcoran et al., 2009; Daro et al., 2011). A LT will allow for optimizing instructional design by offering theory-driven sequences of cognitive progression. The rule space model (RSM; Birenbaum et al., 1993; Tatsuoka, 2009) and attribute hierarchy model (AHM; Leighton et al., 2004) are two earliest CDMs that formally incorporate such a hierarchical structure. Both are pattern recognition approaches rather than a probabilistic measurement model, hence making inferences at item level or attribute level is often difficult. Since then, various methodological studies have incorporated hierarchical relationships into model development (e.g. Bradshaw & Templin, 2014; Gierl et al., 2007; Ma, De La Torre, & Xu, 2023a; Ma, Ouyang, & Xu, 2023b; Tu et al., 2018; Wang, 2024). In addition, many empirical studies have also designed items and assessments to respect certain attribute structures (e.g. Gierl et al., 2007; Wang & Gierl, 2011).

The field of CDMs keeps emphasizing the importance of attribute hierarchy because: (1) cognitive skills, by their nature, do not operate in isolation but belong to a network of interrelated competencies (Kuhn, 2001; Simon & Tzur, 2004; Vosniadou & Brewer, 1992); (2) incorporating information from attribute hierarchy could reduce the number of parameters and improve model fit, which will in turn yield higher classification accuracy of students' attribute profiles (Liu, 2018); and (3) understanding attribute relationships provide explanatory and foundational knowledge related to sequences of cognitive progression (e.g. can several goal facets develop in parallel, or do they develop sequentially?) (Wang, 2024; Wang & Lu, 2021). In this section, we will review both confirmatory and exploratory approaches that either validate extant attribute hierarchies or discover unknown hierarchies from response data. Figure 1 presents a few different types of attribute hierarchy. The arrow indicates prerequisites: for instance, in the convergent relationship, $\boldsymbol{\alpha}_1$ is the prerequisite of $\boldsymbol{\alpha}_2$ and $\boldsymbol{\alpha}_3$, both of which are prerequisites of $\boldsymbol{\alpha}_4$.

2.2.1 | Confirmatory approach

Despite the instructional benefits of a detailed attribute hierarchy map, identifying the multitude of links among attribute pairs poses a methodological challenge. For example, just 9 facets could generate as many as 72 potential directional links. There are two approaches to discover attribute relationships:

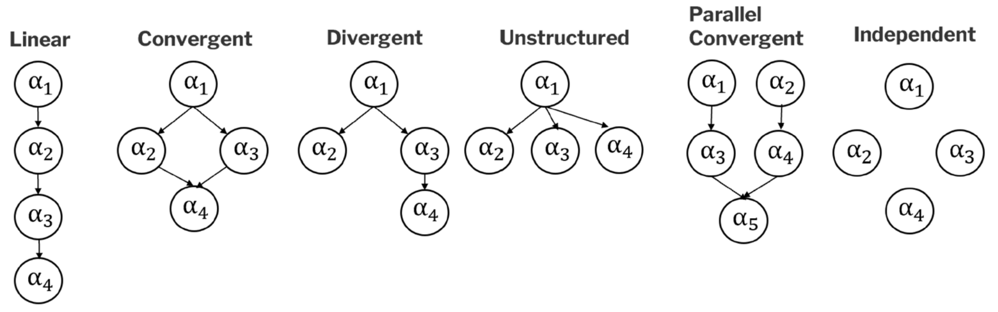

FIGURE 1 Different types of attribute hierarchies.

confirmatory and exploratory approaches. In the confirmatory framework, the attribute hierarchy is specified by experts and then validated by students' response data. For instance, Templin and Bradshaw (2014) proposed a likelihood ratio (LR) test formatted within a hierarchical DCM (HDCM) to statistically test one hypothesized link at a time. That is, they compared two models: one full model assuming independence of two attributes and one reduced model assuming a pre-specified hierarchical relationship between the two attributes. Specifically, if one specifies a saturated CDM for binary responses and two binary attributes (i.e. $\boldsymbol{\alpha}_1$ and $\boldsymbol{\alpha}_2$) as follows:

$$P_j(\boldsymbol{\alpha}_l) = \delta_{j0} + \delta_{j1}\alpha_1 + \delta_{j2}\alpha_2 + \delta_{j12}\alpha_1\alpha_2 \tag{6}$$

Assuming $\boldsymbol{\alpha}_1$ is a prerequisite of $\boldsymbol{\alpha}_2$, according to Templin and Bradshaw (2014), if fitting the model in Equation 6 directly, one would observe a large standard error of $\boldsymbol{\delta}_{j2}$. This is because $\boldsymbol{\delta}_{j2}$ is a simple main effect, which represents the increase in the probability (or log-odds depending on the link function) of a correct response for mastering $\boldsymbol{\alpha}_2$ in the absence of mastering $\boldsymbol{\alpha}_1$. However, in the sample, there will be no one who solely masters $\boldsymbol{\alpha}_2$, hence, there will be no sample for whom this main effect would apply, making $\boldsymbol{\delta}_{j2}$ empirically under-identified. If one fits an additive model instead of the saturated model in Equation 6, then $\boldsymbol{\delta}_{j2}$ will not be empirically under-identified. Instead, $\boldsymbol{\delta}_{j2}$ becomes $\boldsymbol{\delta}_{j12}$ and it implies the increased probability of answering the item correctly when one masters $\boldsymbol{\alpha}_2$ in addition to mastering $\boldsymbol{\alpha}_1$. In addition, instead of estimating the full mixing proportion vector $\boldsymbol{\pi}$, the mixing proportion for profile (0, 1) is 0 due to the hierarchy. Then a LR test is performed to compare the fit of the full model versus the reduced model, thereby evaluating if such a hierarchical link is supported by the empirical data. Although this is theoretically sound, it is cumbersome to perform their LR testing at scale because the reference distribution needs to be simulated each time to derive the empirical p-value. Instead, exploratory approaches that can directly identify massive potential attribute hierarchies from data will not only be computationally preferrable, but also the findings therefrom will complement existing theory-driven LTs and reveal new insights to refine or expand theoretical trajectories.

2.2.2 | Exploratory approach

Several exploratory methods were proposed to identify hierarchical structures directly from data. The first set of methods focuses on recovering the sparse structure of the mixing proportion, $\boldsymbol{\pi}$. With attribute hierarchy, many of 2^K profiles are impermissible, hence, zero elements in $\boldsymbol{\pi}$ suggest the presence of impermissible patterns, whereas non-zero elements correspond to permissible patterns. Wang and Lu (2021) applied latent variable selection (Xu & Shang, 2018) and regularized latent class analysis (Chen et al., 2017) to identify non-zero components in $\boldsymbol{\pi}$. Gu and Xu (2019), as well as Wang (2021, 2024) used a log penalty on $\boldsymbol{\pi}$ in a frequentist expectation–maximization (EM) algorithm to directly shrink insignificant elements in $\boldsymbol{\pi}$ to 0. Once the sparsity structure of $\boldsymbol{\pi}$ is recovered, a heuristic algorithm is used to infer attribute hierarchy

(Wang, 2024). Take $K = 5$ as an example. Assume out of the $2^5 = 32$ potential patterns, only six (i.e. 00000, 10000, 11000, 11100, 1110 and 11111) permissible profiles are identified. Then one can compare pairs of attributes to form adjacency matrix to determine the final hierarchy. For instance, because the first attribute is always mastered whenever any other attribute is mastered, it is deemed the prerequisite for all others. The same pairwise reasoning is applied iteratively to establish the complete hierarchy.

Another way to find sparsity structure of $\boldsymbol{\pi}$ is through hypothesis testing. Liu et al. (2022) proposed a z-statistic to test whether certain mixing proportions (which they term as 'structural parameters') are significantly different from zero. Their z-statistic relies on the estimated standard error, derived from the inverse of the information matrix. However, when some mixing proportions are near-zero, the information matrix can be unstable or even singular. To overcome this numeric instability, Zhang et al. (2024) proposed an iterative extension to iteratively remove structural parameters that are smaller than the pre-specified criterion values.

We need to highlight that the methods reviewed thus far are based on cross-sectional measurement, hence we cannot make direct developmental or causal claims about learning trajectories but only inferring potential attribute hierarchies from snapshots of students' mastery profiles by looking for co-occurrences of masteries (or lack thereof). On the other hand, a suite of latent transition CDMs (TCDM) were proposed in the literature, which is a longitudinal extension of CDMs that model attribute mastery changes over time (Chen et al., 2018; Kaya & Leite, 2017; Li et al., 2016; Madison & Bradshaw, 2018). A shared idea among them is to model the transition probabilities from non-mastery to mastery for each attribute at the group level, also known as the latent transition analysis (LTA: Collins & Lanza, 2010; Collins & Wugalter, 1992; Nylund-Gibson & Hart, 2014). first incorporated attribute hierarchies into the transition probabilities, such as follows: if assuming an attribute-level learning rate of $\boldsymbol{\omega}_k$ and forget rate of $\boldsymbol{\lambda}_k$ for $\boldsymbol{\alpha}_k$, then denote the prerequisites of attribute k as $\{\overline{k}\}$, and denote $\boldsymbol{\alpha}_{ik}^{(t)}$ as person i's mastery status of attribute k at time t, then the latent transition component of a longitudinal CDM becomes,

$$P\left(\boldsymbol{\alpha}_{ik}^{(t+1)} = 1 | \boldsymbol{\alpha}_{ik}^{(t)}, \boldsymbol{\omega}_k, \boldsymbol{\lambda}_k\right) = \begin{cases} 1 - \boldsymbol{\lambda}_k, \text{if } \boldsymbol{\alpha}_{ik}^{(t)} = 1 \\ \boldsymbol{\omega}_k \prod_{k' \in \{\overline{k}\}} \boldsymbol{\alpha}_{ik'}^{(t+1)} \text{ if } \boldsymbol{\alpha}_{ik}^{(t)} = 0 \end{cases} \cdot \tag{7}$$

Apparently, Equation 8 assumes that students learn each attribute independently and it takes a confirmatory approach. In contrast, Wang (2021) proposed an exploratory approach to discover attribute hierarchies from profile-level transitions (Chen et al., 2018; Madison & Bradshaw, 2018). Say if there are L_t and L_{t+1} total number of latent classes at two adjacent time points, then there will be $L_t \times \left(L_{t+1} - 1\right)$ estimable transition probabilities, that is, $\boldsymbol{\tau} = P\left(\boldsymbol{\alpha}^{(t+1)} | \boldsymbol{\alpha}^{(t)}\right)$. This flexible approach will help inform a longitudinal learning progression across different attributes. For instance, the attribute-level transition probabilities, such as $P\left(\boldsymbol{\alpha}_3^{(t+1)} = 1 | \boldsymbol{\alpha}_2^{(t)} = 1\right)$, the transition of mastery of $\boldsymbol{\alpha}_2$ at time t to mastery of $\boldsymbol{\alpha}_3$ at time $(t+1)$, could be summarized from the estimated profile-wise transition probabilities (Madison & Bradshaw, 2018). If $P\left(\boldsymbol{\alpha}_3^{(t+1)} = 1 | \boldsymbol{\alpha}_2^{(t)} = 0\right)$ is close to 0, then $\boldsymbol{\alpha}_2$ is likely a prerequisite of $\boldsymbol{\alpha}_3$. Therefore, when modelling many profile-level transitions, it is likely that some transition probabilities will be close to 0, implying a non-existent pathway. Wang (2021) proposed two novel regularization methods to encourage sparsity of profile-level transitions by shrinking either elements or columns in the transition matrix $\boldsymbol{\tau}$ to 0. The sparsity pattern is then used to infer longitudinal attribute hierarchies, which takes a step further from using cross-sectional data, to understand developmental learning trajectories.

Essentially, these methods reviewed above are 'deterministic' in nature because the sparsity structure of $\boldsymbol{\pi}$ or $\boldsymbol{\tau}$ fully determines the final inferred structure. These methods have two limitations: (1) they only exploit the information of whether a profile is permissible or not but overlook the distributions of attribute profiles. As a result, any error in recovering the sparsity structure of $\boldsymbol{\pi}$ or $\boldsymbol{\tau}$ would severely affect the recovery of attribute hierarchies and (2) they disregard the estimation error in $\boldsymbol{\pi}$ or $\boldsymbol{\tau}$. That is, existing methods demand both low type I error and high power in recovering permissible profiles, conditions that are often difficult to achieve (Liu et al., 2022; Wang & Lu, 2021; Zhang et al., 2024). Instead, two probabilistic approaches recently emerged in literature.

Chen and Wang (2023) proposed a Bayesian Markov chain Monte Carlo (MCMC) algorithm to directly estimate attribute hierarchy. Inspired by structure learning methods in machine learning, the attribute hierarchy can be seen as a directed acyclic graph (DAG, denoted as 'G' in their paper), and their algorithm explored the sample space of DAGs through random walks by adding or removing one directional link at a time until convergence. Their algorithm treats attribute hierarchy as a part of the model parameters and estimates it simultaneously with other CDM parameters. However, their algorithm is model specific, hence, generalizing their algorithm to different types of CDMs needs large efforts. Yan et al. (2024) proposed a heuristic algorithm to infer the hierarchy based on ordering theory (OT). Their method contains two steps. First, a CDM is fitted to the data to estimate individuals' attribute profiles. Then the order among the attributes can be determined based on the sample distribution of the attribute profiles. That is, they compute a K-by-K attribute correlation matrix C, in which $C_{ij} = P\left(\boldsymbol{\alpha}_i = 0, \boldsymbol{\alpha}_j = 1\right)$ is the proportion of individuals who have mastered attribute $\boldsymbol{\alpha}_j$ but not $\boldsymbol{\alpha}_i$. Then a hierarchical index H_i for attribute $\boldsymbol{\alpha}_i$ is a row sum of the matrix C, that is, $H_i = \sum_{j=1}^{K} C_{ij}$. Smaller value of H_i indicates that $\boldsymbol{\alpha}_i$ is at the more basic level of the attribute hierarchy, implying that $\boldsymbol{\alpha}_i$ is more likely the prerequisite of the other attributes. One drawback of their method is that H_i is compared to a somewhat arbitrary cutoff to classify attributes into different levels of the hierarchy. Furthermore, Ren and Wang (2024) showed that their method may not work well when attribute display certain types of structures, such as parallel convergent or independent structures, as shown in Figure 1.

Most recently, Ren and Wang (2025) proposed to use a causal discovery (CD) algorithm to infer attribute relationships. CD algorithms aim to identify the causal relationships among different variables based on a set of observational data (Rodríguez-Nogueira et al., 2022). If viewing the attribute hierarchy as DAGs, the off-the-shelf CD algorithms can be readily applied to either the estimated attribute profiles or the estimated mastery probability matrix from the sample. The sample mastery probability matrix can be obtained from the marginal mastery probability of each attribute for every individual, known as the expected a posteriori (EAP, Huebner et al., 2016). Ren and Wang (2024) demonstrated that a certain attribute hierarchy (e.g. linear hierarchy) is a special case of a strong attribute causal relationship, and while the extant methods reviewed above can recover the strong causal relationship, they do not fare well when there are weak causal relationships among attributes. Furthermore, while the extant CD algorithms can discover the skeleton relationship (i.e. non-directional) among attributes well, they also proposed augmenting the CD algorithm with the OT-based procedure to further assist in discovering *directional* relationships among attributes. Different from Chen and Wang (2023) and like Yan et al. (2024), their method is also a two-stage plug-and-play framework that is compatible with any CDMs. Specifically, in the latent modelling stage, students' profiles are estimated from a specified CDM. Then, in the attribute relationship discovery stage, the hybrid CD-OT algorithm is applied to uncover the attribute relationships.

The hybrid CD-OT algorithm is not without limitations either. First, even though the CD algorithm is used, causal claims cannot be directly made between attributes unless longitudinal data or intervention data is collected. Second, the simple DAG edge structure used in the basic CD algorithm cannot encode 'AND/OR' logic by itself such as $\boldsymbol{\alpha}_3$ requires both $\boldsymbol{\alpha}_1$ AND $\boldsymbol{\alpha}_2$, or $\boldsymbol{\alpha}_3$ requires either $\boldsymbol{\alpha}_1$ OR $\boldsymbol{\alpha}_2$, such as in the convergent structure in Figure 1, which is common in attribute hierarchy. Third, the resulting DAG does not establish strict prerequisites such that hierarchy becomes stochastic. That is, one can only claim mastery of $\boldsymbol{\alpha}_1$ increases the probability of mastering $\boldsymbol{\alpha}_2$ rather than being required. Lastly, the structure in DAGs implies conditional independence due to d-separation, whereas in attribute hierarchies, two attributes may be hierarchically ordered yet still strongly dependent even after conditioning. Hence, cautions need to be exercised when interpreting the findings from the hybrid CD-OT algorithm, and extending the algorithm to longitudinal context would be useful.

2.3 | Fast estimation

The estimation of CDMs can be computationally challenging, especially when the number of attributes is high, due to the 2^K (assuming there are K total number of attributes) possible profiles. This could severely undermine the practical application of CDMs especially if estimation needs to be done on the fly,[2] such as in real time adaptive learning system. Below, we present a few different approaches that can help alleviate the computational challenge.

2.3.1 | Matrix manipulation

We want to start out by highlighting that matrix manipulation is essential to speed up estimation. Although this sounds cliché and may not be psychometrically relevant, we want to share just one concrete example that helps greatly with our own research. The example is quite common in CDM estimations, and the matrix manipulation trick is not often familiar to readers in our field, but the trick is versatile that it can be applied in wide applications, hence worth pointing out here.

Suppose we have a complete-data log-likelihood $\log L_j$ for item j as follows where P_{il} denote the posterior probability of person i in latent class l,

$$\log L_j = \sum_i^N \left[\sum_{l \in \{0,1\}^K} \mathrm{P}_{il} \log \prod_{s=1}^{S_j} \theta_{ls}^{1\left(Y_{ij}=s\right)} \right] \tag{8}$$

In Equation 9, the inner summation over L profiles can be easily done with vector product, and hence a natural programming choice is to create just one loop over N, as shown in Figure 2a and Figure 3 for illustrative MATLAB code. The indicator $1\left(Y_{ij}=s\right)$ obscures direct matrix product between P and $\boldsymbol{\theta}$, but the matrix version overcomes this caveat. Specifically, a matrix version aims to first to conduct a matrix product. Let $\boldsymbol{P}$ denote the $N \times L$ matrix of posterior probability, where N is the number of individuals answering item j and L denotes the total number of latent classes. Let $\boldsymbol{\theta}$ be a $L \times S$ matrix of endorsement probability where S denotes the response categories of item j. Instead of looping through N, we first create a product matrix $\left(\boldsymbol{P} \times \log \boldsymbol{\theta}\right)$, a $N \times S$ matrix. Then

[2]We want to point readers to Nájera, Kreitchmann, et al. (2025) and Nájera, Sorrel, et al. (2025), which provides an innovative approach to variable length computerized adaptive tests that is calibration free. With their method, the on-the-fly calibration challenge can be alleviated.

the indicator function, $1\left(Y_{ij}=s\right)$, translates to select the corresponding column (i.e. determined by the response of person i) for the i th row of the product matrix.

2.3.2 | Reduction of profile space

To further improve computation efficiency, two approaches can be used. The first one is to reduce the number of latent classes from 2^K to only significant latent classes, by penalizing $\boldsymbol{\pi}_l$, as in Gu and Xu (2019), or Wang (2021, 2024). Specifically, a log-penalty can be imposed on $\boldsymbol{\pi}_l$ to encourage its sparsity. That is, with penalty, the objective function that will be maximized takes the following form:

$\arg max_{\boldsymbol{\theta},\boldsymbol{\pi}}\left[logL_N(\boldsymbol{\delta},\boldsymbol{\pi}|\boldsymbol{X})+\gamma\sum_l^{2^K}\log_{\rho_N}\left(\pi_l\right)\right]$ subject to the constraint of $\sum_l^{2^K}\pi_l=1$, where $\boldsymbol{\delta}$ is the model parameters. Here,

$$\log_{\rho_N}\left(\pi_l\right)=\log\left(\pi_l\right)\times I\left(\pi_l>\rho_N\right)+\log\left(\rho_N\right)\times I\left(\pi_l\leq\rho_N\right) \tag{9}$$

where $\rho_N\approx\frac{1}{N}$ is a small threshold parameter to suppress singularity issue of the log function (Gu & Xu, 2019; Wang, 2021). The tuning parameter is $\gamma\epsilon(-\infty,0)$ and smaller γ yields more sparsity in $\boldsymbol{\pi}$. At the convergence of the EM algorithm, $\hat{\pi}_l$ will be compared to ρ_N and only classes with $\hat{\pi}_l>\rho_N$ are retained as permissible classes. Adding this penalty term and carefully selecting the tuning parameter will greatly reduce the latent profile space if the attributes follow certain hierarchical structures. The selection of the tuning parameter can be based on either the BIC or the generalized information criterion (GIC, Gu & Xu, 2019, Wang, 2021, 2024).

The second one stems from Bayesian estimation, namely, the MCMC to be specific. The traditional MCMC method for CDM often uses the Dirichlet distribution as conjugate priors for $\boldsymbol{\pi}$, that is, $\boldsymbol{\pi}\sim$ Dirichlet $(\boldsymbol{\lambda})$ where $\boldsymbol{\lambda}=\left(\lambda_1,\ldots,\lambda_L\right)'$ denotes a $L=2^K$-dimensional hyperparameter vector for $\boldsymbol{\pi}$ (Culpepper, 2015; Culpepper & Hudson, 2018). For large K, this approach often suffers from the increasing computational cost of sampling each latent profile $\boldsymbol{\alpha}_i$ from its full conditional distribution with 2^K categories. This simultaneous Gibbs sampling of the entire $\boldsymbol{\alpha}$ is not only time consuming but also affect the convergence of the MCMC algorithm. Instead, Wang et al. (2022) proposed a sequential Gibbs sampler that aims to sample each attribute one at a time, reducing the computational overhead from $O\left(2^K\right)$ to $O(K)$. The primary idea is to break down the full vector of $\boldsymbol{\alpha}$ into $\{\alpha_k,\boldsymbol{\alpha}_{\backslash k}\}$ where $\boldsymbol{\alpha}_{\backslash k}$ denote a sub-vector that excludes the k th attribute. Then the full conditional distribution of α_k becomes

$$P\left(\alpha_k|\,\boldsymbol{\alpha}_{\backslash k},\boldsymbol{\pi},\boldsymbol{\delta}\right)\propto P\left(\alpha_k|\,\boldsymbol{\alpha}_{\backslash k},\boldsymbol{\pi}\right)P(Y|\,\boldsymbol{\alpha},\boldsymbol{\delta}),$$

where $\boldsymbol{\delta}$ again denotes item parameters. Here, $\left(\alpha_k|\,\boldsymbol{\alpha}_{\backslash k},\boldsymbol{\pi}\right)$ follows a Bernoulli distribution, which implies that the k th attribute will be updated assuming the other attributes are fixed. Hence, the parameter in this Bernoulli distribution can be easily computed as follows:

$$\frac{\sum_{l=1}^{2^K}\pi_l I\left(\alpha_{lk}=1\cap\boldsymbol{\alpha}_{l\backslash k}=\boldsymbol{\alpha}_{\backslash k}\right)}{\sum_{l=1}^{2^K}\pi_l I\left(\boldsymbol{\alpha}_{l\backslash k}=\boldsymbol{\alpha}_{\backslash k}\right)}.$$

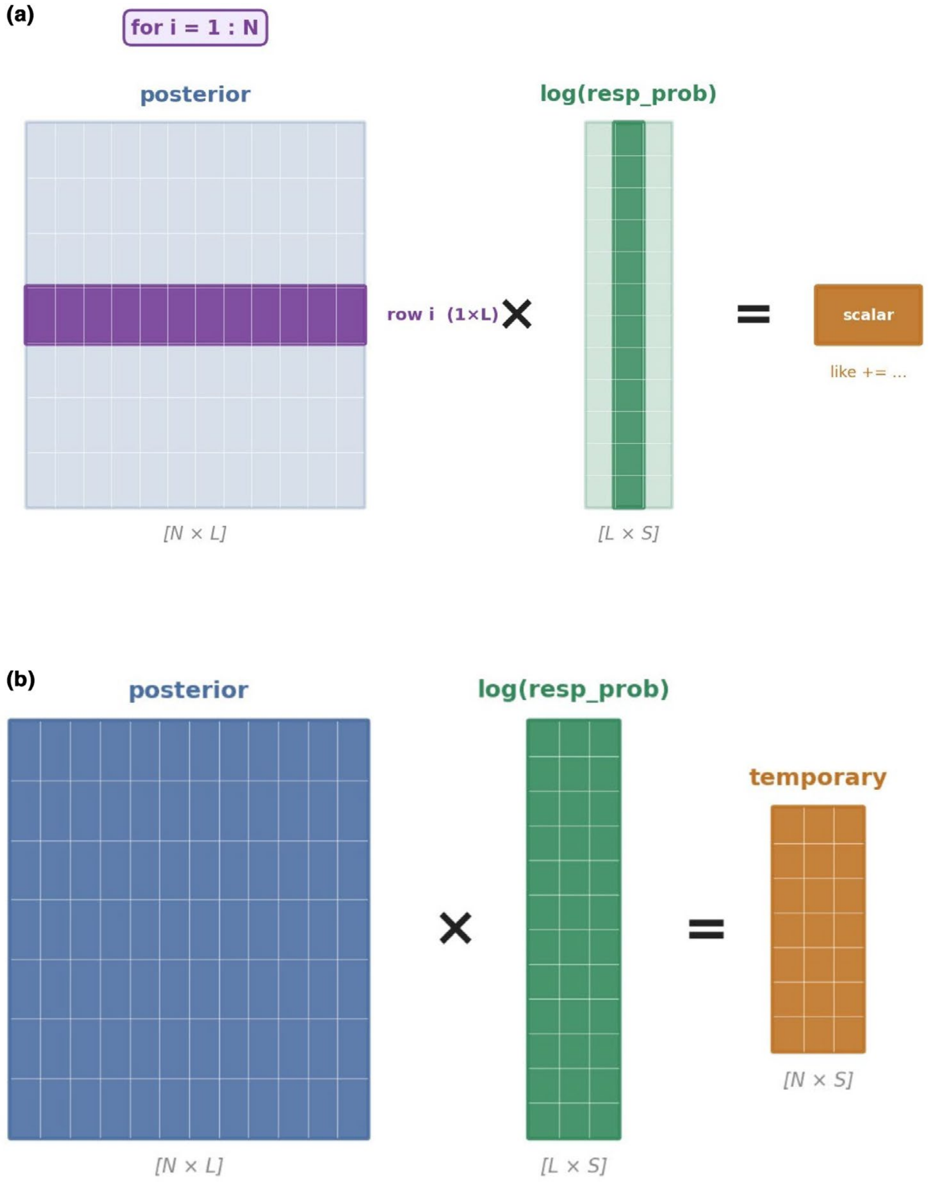

FIGURE 2 Illustration of matrix multiplication. (a) Loop version with vector multiplication (b) Matrix version.

As shown, each element of the attribute profile is sampled separately from the conditional Bernoulli distribution, thereby reducing the computational cost significantly. This approach is suitable for literally any CDMs.

2.3.3 | Two-stage estimation

While having a large number of potential profiles is one cause of slow computation, another possible cause is non-closed-form solution. Taking the DFSM (Wang, 2024) as an example again. In every M-step of the EM algorithm, the item parameters can only be obtained via numeric searches, such as the commonly used L-BFGS-B (Limited Memory Broyden-Fletcher-Goldfarb-Shanno) method.

Loop	
Code	**Comments**
like=0;	initialize a likelihood
for i=1:N	loop over sample size
like=like+posterior(i,:)*log(resp_prob(:,responsej(i)));	"posterior" is a N-by-L matrix that corresponds to "**P**" in Equation 9; "resp_prob" is a L-by-S matrix that corresponds to "**θ**", and "response(i)" corresponds to $1(Y_{ij} = s)$. "response" is a column vector of responses from students who answered item *j*
end	The red highlighted part is a $(1 \times K) \times (K \times 1)$ vector product, and it needs to be done N times.
Matrix	
Code	**Comments**
temporary=posterior*log(resp_prob);	Create a matrix product, $\boldsymbol{P} \times \log \boldsymbol{\theta}$ The red highlighted part is a $(N \times K) \times (K \times S)$ matrix product.
rows=(1:N)'; cols=responsej;	Create indicators to extract relevant elements in the product matrix. "rows" denote "1 to N" for N individuals, and "cols" denote the responses for everyone. The "cols" serves as the indicator in Equation 9, to inform which column of the product matrix needs to be selected for everyone.
like=sum(temporary(sub2ind(size(temporary), rows, cols)));	Sum over all selected elements from the product matrix

FIGURE 3 MATLAB code illustration of Equation 9.

Instead, the G-DINA model or the DINA model enjoy the closed-form solutions for their item parameters. Coupling numeric searches for each item in each M-step with the large number of latent profiles can severely slow down the computation. Instead, a two-step estimation may greatly speed it up.

In Step I, one aims to first estimate θ_{jls} (from Equation 7) for each option s of item j and l th latent profile, using the following closed form,

$$\hat{\theta}_{jls} = \frac{\sum_i P_{il} I\left(Y_{ij} = s\right)}{\sum_i P_{il}}, \tag{10}$$

where P_{il} is the posterior probability of person i belong to profile l. Equation 11 is simply the proportion of individuals who belong to profile l and choose option s of item j.

Although $\boldsymbol{\theta}_{jls}$ is very easy to update, its estimation may not be accurate because there are just too many of them. Several strategies can be used to stabilize the estimation: (1) Instead of estimating $\boldsymbol{\theta}_{jls}$ for all $l \in 2^K$, one can focus only on those $l \in 2^{K_j}$, where K_j is the number of attributes measured by item j. This set can be further reduced during the regularized EM algorithm when some latent classes are deemed impermissible. (2) The initial values matter greatly. It is a known tendency of EM algorithm to converge to local optima when the intital values are poorly chosen (Balakrishnan et al., 2017). Starting all $\boldsymbol{\theta}_{jls}$ at a constant, say one divided by the total number of response options, would lead to the algorithm converges in one cycle, meaning that the estimation gets stuck at a single point. This is because in such a case, the likelihood of every person will be the same simply because the chance of selecting any response option is the same.[3] Although starting $\boldsymbol{\theta}_{jls}$ at a random value will eliminate this issue, but it does not work very well because there are too many parameters and starting randomly will easily move the algorithm towards a local optimum. Instead, informative starting values are necessary. One option could be to use nonparametric methods (see Section 2.4) to first roughly classify everyone into one of the latent classes. Then compute the proportion of individuals in each latent class who select option s of item j. This treatment resembles closely to Equation 11 closely, except that it does not use the posterior probability P_{il}.

In Step II, given $\hat{\boldsymbol{\theta}}_{jls}$ and $\hat{\boldsymbol{\pi}}_l$ from step I, we can obtain item parameters for each item separately by minimizing the following loss function

$$Loss_j = \sum_l \hat{\pi}_l \left(\sum_s \left| \hat{\theta}_{jls} - \tilde{\theta}_{jls} \right| \right). \tag{11}$$

In Equation 12, $\tilde{\boldsymbol{\theta}}_{jls}$ is the predicted probability from any CDM that includes relevant item parameters one aim to estimate. The absolute value in Equation 12 can be changed to other forms such as squared difference, and the estimation precision will not change too much. We assume Q-matrix is known in this case.

This two-step approach separates the computationally intensive search of optimal model parameters (step II, Equation 12) from the easy update of $\hat{\boldsymbol{\theta}}_{jls}$ from step I. As described above, $\hat{\boldsymbol{\theta}}_{jls}$'s are obtained through the EM algorithm, and the model parameters are updated outside of the EM algorithm as a separate step. This is different from the EM algorithm of G-DINA (De La Torre, 2011), which also includes an implicit two-step update in the M-step. In the G-DINA EM algorithm, the two-step refers to updating $\hat{\boldsymbol{\theta}}_{jls}$ first and updating model parameters as linear combinations of $\hat{\boldsymbol{\theta}}_{jls}$ in every M-step *within* the EM cycle. Hence, the two-step updates are inside the EM cycle. Although separating the two-step updates outside of the EM cycle greatly speeds up computation, their stability and performance need to be further evaluated. This is because the successful updates in Step II hinge on precise estimates of $\hat{\boldsymbol{\theta}}_{jls}$ from step I, which can be challenging due to the sheer number of parameters that need to be updated simultaneously. Our limited pilot exploration reveals that although step I can converge quickly, it may sometimes converge to local optima and adversely affect step II estimation accuracy. Hence, more thorough studies and maybe some adjustments to the two-step approach will be needed to ensure its overall accuracy is on par with the classic one-step approach. Inspired by Sorrel et al. (2017), who proposed a two-step likelihood ratio test which shares a similar spirit to the two-step approach, we can make some adjustments to the general-purpose two-step approach. Specifically, in Sorrel et al. (2017), they proposed to estimate the G-DINA model first, and then in the second step, the item parameters from the reduced models can be obtained using $\hat{\boldsymbol{P}}_j$ (like $\hat{\boldsymbol{\theta}}_{jls}$ from Equation 11) from the G-DINA model. As a result, the item parameters from the reduced models are therefore obtained from a two-step approach. The key is how $\hat{\boldsymbol{P}}_j$ is computed: in the general stated two-step approach above, this statistic is obtained from EM algorithm independent of any CDMs, whereas in Sorrel et al. (2017), $\hat{\boldsymbol{P}}_j$ is obtained as a by-product from the EM algorithm for G-DINA. In this regard, their $\hat{\boldsymbol{P}}_j$'s will be more accurate compared

[3]Regarding the initial values of $\boldsymbol{\pi}_l$, we recommend random start instead of a constant of $\frac{1}{2^K}$ for the same reason.

to $\widehat{\boldsymbol{\theta}}_{jls}$. Hence an adjustment can be made as follows: In stage I, one can first construct a general model (e.g. sp-G-DINA, De La Torre et al., 2025; or Ma, 2022) that has closed-form updates of item parameters from $\widehat{\boldsymbol{\theta}}_{jls}$ (or $\widehat{\boldsymbol{P}}_j$), and the goal is to calculate $\widehat{\boldsymbol{\theta}}_{jls}$ (or $\widehat{\boldsymbol{P}}_j$) as a byproduct from model estimation. This way, $\widehat{\boldsymbol{\theta}}_{jls}$ (or $\widehat{\boldsymbol{P}}_j$) will be more accurately obtained than estimating them standalone because in the latter case, the number of parameters is much higher. Then, in stage II, use $\widehat{\boldsymbol{\theta}}_{jls}$ (or $\widehat{\boldsymbol{P}}_j$) to estimate other reduced model parameters that do not have closed-form solutions.

In sum, the methods reviewed above are general-purpose approaches that can be applied to different types of CDMs. This section review is by no means exhaustive, and there are other theoretical and algorithmic advancements that make CDM estimation computationally more affordable for specific types of CDMs. For instance, Culpepper (2015) proposed a Gibbs sampling algorithm for the DINA model, and Culpepper and Hudson (2018) proposed a Gibbs sampling algorithm for the reduced reparametrized unified model (rRUM). Both studies show that the improved strategy for Bayesian estimation converges faster than the commonly used Metropolis-Hastings algorithm, thereby reducing the computation time. Zeng et al. (2023) proposed a tensor-EM algorithm to handle CDM estimation with binary responses in the double asymptotic regime, that is, when both sample size and test length go to infinity. Their method contains two steps: using a moment-based tensor power method in the first step and then using the obtained estimates as initialization for the EM algorithm in the second step. Not only does the two-step method achieve good accuracy and computational efficiency, but it also provides theoretical guarantees of the clustering consistency in this double asymptotic regime.

2.4 | Challenges with small sample sizes

Most classrooms yield relatively small datasets, raising concerns about whether CDMs can be reliably implemented in such settings. Sessoms and Henson (2018) reviewed a range of CDM applications and found that 61% involved sample sizes greater than 1000, with another 31% falling between 1000 and 2000. Only two studies—Im and Yin (2009) and Jang et al. (2015)—used sample sizes comparable to those typically found in classroom contexts, which, for the purposes of this review, we define as ranging from approximately 20–100 students. Such sample sizes are common in individual courses, small instructional programmes, or pilot evaluations, all of which could benefit from the use of CDMs.

In this section, we first provide an overview of challenges that small sample sizes pose for CDMs, including (1) model estimation, (2) model assessment and selection and (3) Q-matrix validation/estimation. We then discuss how some of these challenges are overcome by nonparametric cognitive diagnostic methods. Finally, we discuss connections between parametric and nonparametric methods, with a particular focus on how Bayesian methods have the potential to overcome many of the challenges that arise in small-sample cognitive diagnostic modelling.

2.4.1 | Small sample estimation of CDMs

A consistent finding in the literature is that larger sample sizes lead to better estimation properties. For instance, De La Torre et al. (2010) showed that increasing sample size from 1000 to 4000 in the DINA model reduced bias and RMSE. Similar patterns hold for DINO, A-CDM and G-DINA models (Rojas et al., 2012), with larger samples improving attribute classification accuracy. Kunina-Habenicht et al. (2012) found that sample sizes of 1000 were sufficient for estimating main effects in the LCDM, but that sample sizes of up to 10,000 were needed to estimate higher-order interaction effects. Studies confirm these patterns across CDMs such as C-RUM (Galeshi & Skaggs, 2016), DINA (Sünbül &

Kan, 2016) and G-DINA (Başokçu, 2014). Sen and Cohen (2021) concluded that samples of at least 500 are needed for reliable estimation in simpler models, with more complex models requiring larger samples.

Estimation challenges in small samples are not limited to degraded accuracy—they can include complete estimation failure. When estimating the DINA model via the EM algorithm, DeCarlo (2011) observed boundary estimates and proposed a Bayesian solution. Templin and Bradshaw (2014) reported frequent non-convergence for hierarchical models with small samples; Ma and Guo (2019) and Oka and Okada (2021) documented similar issues. Bayesian methods can also be unstable with uninformative priors in small samples (Oka & Okada, 2021).

As a simple example, consider the estimator for slipping parameters in the DINA model:

$$\hat{s}_j = \frac{\sum_{i=1}^{N}\left(1-y_{ij}\right)\eta_{ij}}{\sum_{i=1}^{N}\eta_{ij}}, \tag{12}$$

where $\eta_{ij} = 1$ if individual i has all attributes required by item j. Boundary estimates occur if all mastery-group respondents answer correctly or incorrectly. If no one is in the mastery group, $\sum_{i=1}^{N}\eta_{ij} = 0$, and the estimate is undefined. Even without these extremes, small N causes high variance due to the small denominator.

Sample size also influences model fit indices and selection. Choi et al. (2010) showed that log-linear fit indices require at least 200 samples for stability, with reliability improving further at larger samples (Hu et al., 2016; Lei & Li, 2016). Model selection tests, such as Wald or likelihood ratio tests comparing G-DINA to reduced models, need large samples (often >2000) to maintain Type I error control and selection accuracy (De La Torre & Lee, 2013; Sorrel et al., 2021).

In many applications, the Q-matrix may be misspecified or entirely unknown and must therefore be validated or estimated. This process is also sensitive to sample size. Liu et al. (2012) demonstrated that Q-matrix recovery probability increased markedly as sample size rose from 500 to 4000. Similar trends appear across R-RUM, DINO and G-DINA models and hold under both frequentist (Chen et al., 2015; Chung, 2019; Xu & Shang, 2018) and Bayesian estimation (Chen et al., 2015, 2018; Culpepper, 2019).

These findings reveal how sample size impacts every CDM component—parameter estimation, fit assessment, classification, model selection and Q-matrix validation. Nevertheless, the classroom remains a central context for formative assessment and understanding how to adapt CDMs to small-sample settings is essential for extending their practical utility. This need has motivated research into methods that can work more effectively in these settings.

2.4.2 | Clustering and nonparametric cognitive diagnosis

In response to the limitations of traditional CDMs in small sample settings, early researchers developed nonparametric methods that offer more robust classification performance—particularly with respect to accuracy. These methods often avoid the need for full parametric estimation of item parameters, which is unstable or infeasible when sample sizes are small.

Early work applied clustering heuristics assuming respondents with similar mastery patterns yield similar response profiles. Willse et al. (2007) used k-means clustering with the R-RUM model; Ayers et al. (2008) and Chiu et al. (2009) showed that k-means and hierarchical clustering consistently recover attribute profiles as item count grows. Neural network classifiers were tested (Shu et al., 2013) but proved more sensitive to item quality and noise.

These successes set the stage for formal nonparametric CDMs, the most studied small-sample alternatives to parametric models. Chiu and Köhn (2019) introduced a DINA-based method for assigning attribute profiles by minimizing a weighted discrepancy between observed and ideal response patterns:

$$\hat{\alpha}_i = \underset{c}{\operatorname{argmin}} \sum_{j=1}^{J} \frac{1}{\hat{p}_j\left(1-\hat{p}_j\right)} \left|y_{ij} - \eta_{jc}\right|, \tag{13}$$

where $\hat{p}_j$ is the proportion correct for item j and η_{jc} is the ideal response under attribute profile c, assuming a conjunctive response function. This method achieves consistent classification as the number of items increases (Chiu & Köhn, 2015), with extensions to DINO, NIDA and R-RUM. Furthermore, several simulations have shown that nonparametric approaches outperform maximum likelihood estimation in small samples (Akbay, 2016; Paulsen & Valdivia, 2022; Tzou & Yang, 2019).

While early nonparametric methods focused on DINA's strict conjunctive structure, Chiu et al. (2018) generalized them to the G-DINA model. In this approach, rather than letting ideal responses conform to a single model, the ideal response is a weighted average of conjunctive and disjunctive components:

$$\eta_{jc} = w_{jc}\eta_{jc}^{c} + \left(1 - w_{jc}\right)\eta_{jc}^{d}, \tag{14}$$

where η_{jc}^{c} and η_{jc}^{d} represent the ideal responses under the conjunctive and disjunctive models, respectively, and w_{jc} is a data-driven weight estimated from item responses. This method also has proven consistency (Chiu & Köhn 2019) and outperforms parametric models in many settings. Nonparametric cognitive diagnosis has since been extended to other tasks, including Q-matrix refinement (Chiu, 2013) and computerized adaptive testing (Chang et al., 2019; Chiu & Chang, 2021), making it a versatile and practical approach for diagnostic assessment in low-data settings. Together, these developments highlight how nonparametric approaches provide both statistically principled and computationally efficient solutions for cognitive diagnosis when sample sizes are severely limited.

2.4.3 | Parametric vs. nonparametric methods

While nonparametric methods have proven highly effective for cognitive diagnosis in small sample settings—especially in terms of attribute classification accuracy—they are not without limitations. We highlight four main challenges.

First, nonparametric methods typically lack built-in mechanisms for quantifying uncertainty in classifications. Second, although estimating item parameters in small samples is difficult, such parameters are still valuable for practitioners aiming to evaluate item and test quality. Third, nonparametric methods do not provide a natural framework for model comparison, making it difficult to evaluate competing CDMs. Fourth, unlike parametric models, increasing the sample size in nonparametric methods does not guarantee improved performance, raising practical questions about when to use one approach over the other.

Recent research reveals connections bridging parametric and nonparametric CDMs. Ma et al. (2023) proposed a unified framework where parametric models (DINA, G-DINA) and nonparametric classifiers (NPC, GNPC) arise by varying a general loss function over item response–centroid discrepancies. This framework bridges the conceptual gap between the two modelling paradigms and suggests that hybrid approaches are feasible (see Yamaguchi et al., 2025 for a Bayesian extension).

Nájera et al. (2023) introduced the restricted DINA (R-DINA) model, which is a simplified version of the DINA model that assumes $s_j = g_j = \phi$. That is, all item parameters are replaced by a single parameter for all items. They showed that MAP estimates from this model match NPC estimates while also allowing for uncertainty quantification and model comparison (see Sorrel et al., 2023 for practical recommendations regarding implementation in classroom assessment). The parameter ϕ in the R-DINA model is as follows:

$$\hat{\phi} = \frac{\sum_{i=1}^{N} \sum_{j=1}^{J} y_{ij} \eta_{ij}}{\sum_{i=1}^{N} \sum_{j=1}^{J} \eta_{ij}}. \tag{15}$$

In comparison to the earlier estimator for s_j, we can see that this is much less likely to exhibit problems with boundary or undefined estimates, even when N is small. It will also exhibit lower variance, at the cost of introducing more bias as an estimator of s_j.

These results highlight a bias-variance tradeoff with CDMs: rich parametric models (e.g. G-DINA) have low bias but high variance in small samples; highly constrained models (e.g. R-DINA) have low variance but higher bias. Nonparametric methods fall on the low-variance end but lack the flexibility of full parametric estimation. Ideally, one would aim for a balance between these extremes.

Bayesian estimation offers a promising resolution. Ma and Jiang (2021) showed that shrinkage priors improve classification and parameter stability in G-DINA. Using slightly informative Beta priors—Beta (1.5, 2.5) for slipping/guessing and Beta (2, 2) for others—they obtained posterior mode estimates via Bayes modal estimation. These priors shrank item parameters towards interpretable values (e.g. .25 or .5), reducing variance with minimal bias. Bayesian modal estimation modifies the DINA slipping estimator as follows:

$$\hat{s}_j = \frac{\sum_{i=1}^{N} \left(1 - y_{ij}\right) \eta_{ij} + .5}{\sum_{i=1}^{N} \eta_{ij} + 2}. \tag{16}$$

Using a prior distribution, the risk of boundary or undefined estimates is eliminated. Like the R-DINA model, this is done by introducing bias and reducing variance. However, in contrast to the R-DINA model, the reduction in variance is slight and the bias minimal.

Each of these methods involves fixed priors or fitting more restrictive models. A more flexible approach would involve using a Bayesian hierarchical model. For example, the following illustrates a hierarchical prior for the DINA model slipping parameters:

$$s_j \sim \text{Beta}(\tau s, \tau(1-s)), s \sim \text{Beta}(1,1), \tau \sim \text{Gamma}(a,b).$$

These priors act as shrinkage priors, with the amount of shrinkage being controlled by τ. If $\tau = 4$ and $s = .375$, this reduces to Bayes modal estimation. As $\tau \to \infty$, the model behaves more like the R-DINA model. The hierarchical model represents a compromise between these two extremes. The benefits of such an approach were illustrated by Arthur (2024). They used an Empirical Bayes approach to hierarchical modelling for the DINA model via catalytic priors (Huang et al., 2020, 2025). Not only did this method outperform both NPC and Bayes modal estimation in small-sample simulations, but performance continued to improve as N and J increased.

These recent developments suggest that the best approaches for estimation and classification involve restricting complex CDMs or using Bayesian methods to shrink complex models toward simpler models with desirable properties. These findings might provide insights into other areas of small-sample cognitive diagnostic modelling that have received less attention. For example, Bayesian model selection methods have shown some promise in small-sample model selection tasks (Arthur, 2024). Several approaches also exist for Q-matrix estimation, with many of these being Bayesian methods. However, little research has been done regarding how these methods might be adapted to work better in smaller sample settings. The principles of parsimony and shrinkage could prove useful in both of these areas of research and could help make CDMs more accessible in more educational assessment settings.

2.5 | Q-matrix estimation and refinement

Q-matrix is a critical component of CDMs because not only its design determine whether a specific CDM is statistically identifiable (known as completeness, Köhn & Chiu, 2018), but also it reveals the coverage of attributes among items and hence affects whether individuals' latent profiles can be precisely recovered (Madison & Bradshaw, 2015). Ideally, Q-matrix should be created prospectively when items are generated by context experts. These initial Q-matrix can be validated and refined using both quantitative and qualitative data, such as students' responses or their cognitive interview narratives. Oftentimes, however, we do not have access to such data from beginning to end. Instead, we have partial data, such as only students' responses without an expert-defined Q-matrix, or experts may have divergent perspectives on candidate Q-matrices. These practical scenarios motivate the development of data-driven methods, which can be divided into two sets: methods for *refining* and *validating* Q-matrix when expert-defined Q-matrix are available, but it may exhibit different versions, and methods for *estimating* Q-matrix when no initial Q-matrix is available. We need to emphasize that obtaining the optimal Q-matrix is an iterative process, and despite the statistical methods, experts still need to be in the loop during the process to ensure Q-matrix validity.

2.5.1 | Q-matrix refinement

Refinement is also considered as Q-matrix validation. It is used when one already has partial knowledge of the Q-matrix and seek to refine and validate every entry or a certain proportion of the entries of a given Q-matrix. The main idea of most refinement methods is to separate the unknown true q-vector for item j from all $2^K - 1$ possible candidate q-vectors for each item. Such a separation can be quantified by different statistical indices. For instance, Chiu (2013) proposed a refinement method for conjunctive CDMs like the DINA model. For the DINA model, an ideal item response for individual i to item j is $\eta_{ij} = \prod_{k=1}^{K} \alpha_{ik}^{q_{jk}}$ for a given $\boldsymbol{q}_j$. Then, among all possible $\boldsymbol{q}_j$'s, the true $\boldsymbol{q}_j$ should be the one that makes the ideal responses closest to the observed responses. Hence, Chiu (2013) defined a residual sum of squares (RSS) as $RSS_j = \sum_{i=1}^{N} \left(y_{ij} - \eta_{ij}\right)^2$ and proved that the expectation of RSS_j is minimized for the true $\boldsymbol{q}_j$. While Chiu (2013)'s method only applies to conjunctive CDMs, De La Torre and Chiu (2016) proposed a G-DINA model discrimination index (GDI) that applies to general CDMs. The GDI is a generalization of De La Torre's (2008) $\delta-$

method, and it purports to select $\boldsymbol{q}_j$ that maximizes the weighted variance of the correct response probabilities:

$$\varsigma_j^2 = \sum_{l=1}^{2^K} P(\boldsymbol{\alpha}_l)\left[P\left(Y_j = 1|\boldsymbol{\alpha}_l\right) - \overline{P}_j\right]^2 \tag{17}$$

where $P(\boldsymbol{\alpha}_l)$ is the posterior probability for attribute profile l, and $\overline{P}_j = \sum_{l=1}^{2^K} P(\boldsymbol{\alpha}_l)P\left(Y_j = 1|\,\boldsymbol{\alpha}_l\right)$ is the weighted average. They showed that the correct $\boldsymbol{q}_j$ should maximally differentiate success probabilities across all relevant latent classes. Most recently, Li and Chen (2025) introduced a β index that approaches Q-matrix validation as a signal detection task: separating the true signal (i.e. the unknown true $\boldsymbol{q}_j$) from noise (i.e. all $2^K - 1$ possible candidate q-vectors). The rationale is that among all candidate q-vectors, the true $\boldsymbol{q}_j$ provides highest overall discrimination, resulting in the optimal decision outcome—highest difference between hit rate (denoted as H) and false detection (denoted as F). Taking the DINA model as an example again. By treating the observed responses to item j as signal, the hit rate is the proportion of individuals belong to latent class l who answer item j correctly, that is, $H = \frac{r_{lj}}{n_l}$, where n_l is the number of individuals in class l, and r_{lj} is the number of individuals in class l who answers item j correctly. Treating the model expected responses using a given $\boldsymbol{q}_j$ as noise, the false detection is $F = 1 - P\left(Y_j = 1|\,\boldsymbol{\alpha}_l\right) \equiv 1 - P_{jl}$. The β index is therefore defined as

$$\beta = \sum_{l=1}^{2^K}\left|\frac{r_{lj}}{n_l} - \left(1 - P_{jl}\right)\right| = \sum_{l=1}^{2^K}\left|\frac{r_{lj}}{n_l}P_{jl} - \left(1 - \frac{r_{lj}}{n_l}\right)\left(1 - P_{jl}\right)\right|. \tag{18}$$

Equation 22 also makes intuitive sense because $\frac{r_{lj}}{n_l}P_{jl}$ can be interpreted as the proportion of individuals with both observed and expected responses of 1, and the second product is the proportion of individuals with both observed and expected responses of 0. It is easy to note that the larger the first product is, the smaller the second product is. If the observed and expected responses closely align, one would expect the two product terms to be quite different, resulting in a large $\boldsymbol{\beta}$. Hence, the optimal $\boldsymbol{q}_j$ should maximize $\boldsymbol{\beta}$.

Note that Equations 19 and 22 are different ways of quantifying what one would observe had we known true $\boldsymbol{q}_j$. They are considered refinement and validation methods because they need a provisional Q-matrix to obtain estimates of item parameters and posterior distribution of the attribute profiles. Hence, it is essentially a two-step procedure. In contrast, Culpepper (2019) proposed a Bayesian estimation of Q-matrix using a prior based on expert knowledge and considered a fully Bayesian formulation for a general diagnostic model. His method uses a novel prior that incorporates expert knowledge as predictors within a latent, multivariate regression model. Specifically, suppose $\boldsymbol{X}$ is a $J \times (V + 1)$ design matrix with the first column of 1's to denote intercept. $\boldsymbol{\Gamma}$ is a $(V + 1) \times K$ regression coefficient that directly relate expert knowledge to the Q-matrix, such that $q_{jk} \sim \text{Bernoulli}\left[\Phi\left(\boldsymbol{x}_j'\boldsymbol{\gamma}_k\right)\right]$. Culpepper (2019) claimed that $\mathbf{\Gamma}$ should be sparse because a predictor should relate to at most one attribute; hence, he proposed a spike-and-slab prior on $\mathbf{\Gamma}$. The advantage is one may uncover new, unanticipated attributes that can help advance cognitive theory.

Another approach to Q-matrix refinement is through item fit because a misspecified q-vector for an item may lead to item misfit. Popular methods include the General Discrimination Index (GDI; De La Torre & Chiu, 2016), which examines how well items discriminate between different latent classes, and residual-based analysis approaches (Chen, 2017) that focus on systematic patterns in model residuals to identify Q-matrix misspecifications. The convex hull approach (Ceulemans & Kiers, 2006; Nájera et al., 2021) offers another perspective by examining whether the empirical item response patterns fall within the theoretical boundaries defined by the Q-matrix structure. This geometric approach can be particularly useful for identifying items that may require additional attributes not currently specified in the Q-matrix.

Beyond these classic methods, current cutting-edge item-fit methods developed for Q-matrix refinement have increasingly focused on applying machine learning (ML) and artificial intelligence (e.g. Qin & Guo, 2023). For example, Qin and Guo address the limitations of the GDI and the Hull methods by proposing four new Q-matrix item-fit methods that utilize feed-forward neural networks and random forests. They note that the GDI method relies on a predetermined cutoff point (e.g. .95), which can be difficult to determine and may not be robust across different data conditions, and the Hull method tends to under-specify the Q-matrix by rarely selecting the q-vector where all attributes are measured by an item. Rather than approaching the item-fit problem from a statistical test viewpoint, Qin and Guo propose that finding the most appropriate q-vector from all possible q-vectors is analogous to a classification task in machine learning.

Specifically, the random forests and feed-forward neural networks are trained using the proportion of variance accounted for (PVAF) and the McFadden pseudo-R^2 calculated from labelled simulated datasets as features (i.e. input). For a given item j and a candidate q-vector c, we can calculate the PVAF as

$$\mathrm{PVAF}_{jc} = \frac{\xi_{jc}^2}{\xi_{1:K}^2} \tag{19}$$

where ξ_{jc}^2 is the weighted variance of the probabilities of correctly responding to item j under candidate q-vector c across all attribute profiles relevant to item j (i.e. GDI, De La Torre & Chiu, 2016), and $\xi_{1:K}^2$ is the maximum possible value which occurs when we have a true q-vector. Therefore, there will be $2^K - 1$ different values of PVAF_{jc} for item j as its inputs to train its q-vector. Similarly, the McFadden pseudo-R^2 can be calculated as

$$\mathrm{R}_{jc}^2 = 1 - \frac{log\left(L_{jm}\right)}{log\left(L_{j0}\right)} \tag{20}$$

where $log\left(L_{jm}\right)$ is the log-likelihood of a specified CDM model, and $log\left(L_{j0}\right)$ is the log-likelihood of the null model. Here, the null model computes the 'correct response probability' as the average probability of correctly responding to an item across all N individuals.

Once the models were trained, their performance was evaluated on simulated labelled testing data. The study conducts two simulation studies and a real-data analysis to evaluate the new ML-based methods against the GDI and Hull methods. The evaluation criteria include the Q-matrix recovery rate (QRR), vector recovery ratio (VRR), under-specification rate (USR) and overspecification rate (OSR). QRR is the proportion of correctly specified q-entries, and VRR is the ratio of correctly recovered q-vectors to the total number of items. USR and OSR measure the tendency to under-specify or over-specify the Q-matrix, respectively, where USR + OSR + QRR = 1. High values of QRR suggest that the method was able to correctly recover the data-generating Q-matrix. The results of the simulation studies show that the ML-based methods generally outperform the GDI and Hull methods. The GDI method had the lowest QRR and VRR values and elevated OSR values, which suggest a tendency to over-specify the Q-matrix. The Hull method produced the largest USR values across all methods and tended to under-specify the Q-matrix. In contrast, the ML-based methods balanced under- and overspecification and showed significantly higher QRR and VRR values than the GDI and Hull.

2.5.2 | Q-matrix estimation

Q-matrix estimation refers to directly estimating the entire Q-matrix from data. Due to space limit, this review omits the extensive work around the identifiability of Q-matrix, which is critical in deriving data-driven Q-matrix. Interested readers should refer to Zhang et al. (2024) and references cited therein.

The estimation methods generally fall into three categories: exploratory factor analysis (EFA), regularization within frequentist approach, and Bayesian methods. The EFA method starts with a tetrachoric inter-item correlation matrix. Then, given K factors, the rows of the $J \times K$ item loading matrix form the basis of the Q-matrix. For a given item j, first ordering the factor loadings in ascending order, then the proportion of the communality of item j accounted for by the consecutive sum of the first several factors is compared to a pre-specified cutoff to decide on the q-vector (Köhn et al., 2025). The cutoff is chosen to yield the best model-data fit. For the regularization method, Xu and Shang (2018) developed an EM algorithm for likelihood-based estimation of the Q-matrix with truncated $L1$ penalty. Their method can be used with any CDMs, in that the $L1$ penalty is imposed on the regression coefficients (i.e. $\boldsymbol{\delta}_j$) hence it selects significant $\boldsymbol{\delta}_j$'s that implies the relevant attributes measured by item j. Lastly, within the Bayesian framework, Chen et al. (2018) and Liu et al. (2020) developed MCMC algorithms for Q-matrix estimation for the DINA model, with Chen et al.'s (2015) identifiability constraints implemented as constraints. Culpepper and Chen (2019) used a spike-slab prior for item parameters to select the required attributes for each item, thereby estimating the Q-matrix for the reduced reparametrized unified model (rRUM).

2.6 | Model fit evaluation

CDM model fit evaluation is essential for ensuring the validity of the estimated latent attribute profiles. For this section, we review both global model fit evaluation methods and methods for evaluating item fit. Table 3 presents a summary of extant global model fit evaluation methods and item fit tests.

In the CDM literature, misfit in CDMs stems from two major sources: model misspecification and Q-matrix misspecification. CDM model misspecification occurs when an incorrect model is used to represent the relationship between latent attributes and item responses. For instance, a researcher may apply a conjunctive model, such as the DINA, when the true data-generating process is disjunctive, which is better captured by a model like the DINO. CDM misspecification can result in inaccurate classification of individuals' latent attribute mastery profiles, even when the Q-matrix is correctly specified.

Because different CDMs apply different assumptions about how attributes combine to produce item responses, the selection of an incorrect model can compromise both the diagnostic validity and the substantive interpretability of the results.

Q-Matrix misspecification generally follows one of four patterns: overspecification, under specification, balanced misspecification or attribute creation/reduction (e.g. Chen, 2017; Lei & Li, 2016; Liu et al., 2019; Rupp & Templin, 2008). An overspecified Q-matrix occurs when an attribute that is not required for mastery is misspecified as being required (i.e. specifying a 1 where a 0 should be present). Conversely, an underspecified Q-matrix occurs when an attribute that is required for mastery is misspecified as not required (i.e. specifying a 0 where a 1 should be present). Balanced misspecification involves both types of errors within the same Q-matrix where some required attributes incorrectly specified as 0 and some non-required attributes incorrectly specified as 1. Finally, attribute creation/reduction involves either adding an unnecessary latent attribute or omitting a required latent attribute from the model entirely. Since we have reviewed the latest Q-matrix refinement and validation methods in Section 2.5, in this section, we focus on general item fit evaluation methods.

2.6.1 | Global model fit evaluation

The goal of global CDM model fit evaluation is to identify the model that best accounts for the observed response patterns. This process involves comparing both relative and absolute model fit indices among a pool of candidate CDMs (Chen et al., 2013). Since the true data-generating mechanism remains unknown, researchers must rely on empirical fit indices to guide model selection.

Relative model fit focuses on selecting the best-fitting CDM among candidate models, while absolute model fit investigates whether a given model fits the data adequately in an absolute sense. Examples of relative fit indices include the Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC), where smaller values indicate better fit. *Absolute model fit* investigates whether a given model fits the data adequately without comparison to other models. Absolute fit indices include the Root Mean Square Error of Approximation (RMSEA), the Mean Absolute Deviation of item correlations (MADcor; DiBello et al., 2006) and traditional chi-square tests such as Pearson's χ^2 and the likelihood ratio G^2 test (Han & Johnson, 2019).

Chen et al. (2013) proposed that examining relative and absolute fit statistics sequentially provides the most comprehensive picture of global model fit. They recommended that researchers first use relative fit indices to identify the best-fitting models, then examine absolute fit indices to detect potential Q-matrix misspecifications between the models. Their research demonstrated that relative fit indices are particularly sensitive to CDM model misspecification, while absolute fit indices are more effective at detecting Q-matrix misspecification. This work was extended by Hu et al. (2016) and Hansen et al. (2016), who examined the performance of additional absolute fit indices for detecting Q-matrix misspecification. These indices include the limited information RMSEA2 (Houts & Cai, 2013; Maydeu-Olivares & Joe, 2014), the absolute value of the deviations of Fisher-transformed correlations (ABS(fcor); Chen et al., 2013; Robitzsch et al., 2014), and the maximum of all item pair chi-square statistics (MAX($\chi^2_{jj'}$), Robitzsch et al., 2014).

Under ideal conditions, relative and absolute fit indices should converge on the same conclusions about model adequacy. However, in practice, researchers may encounter situations where relative and absolute fit indices provide somewhat inconsistent information, particularly when Q-matrix misspecification affects different items to varying degrees. For example, Li et al. (2016) examined five candidate CDMs for modelling the Michigan English Language Assessment Battery. The relative fit indices suggested that the G-DINA fit the data best, followed by ACDM, or R-RUM, while absolute fit indices indicated that ACDM and G-DINA provided equally good fit, whereas R-RUM displayed a slightly worse fit. This finding illustrates a common challenge in model selection: while the relative indices suggested G-DINA best fit the data, the absolute indices indicated that both G-DINA and ACDM were

TABLE 3 CDM model and item fit tests.

Fit category	Method/Index	Description & purpose
Relative model fit	−2 Log-Likelihood (−2LL)	The model with the lowest-2LL is preferred. Tends to prefer a fully saturated model.
	Akaike information criterion (AIC)	An information-theoretic statistic used to compare non-nested models. The model with the lower AIC value is preferred.
	Bayesian information criterion (BIC)	Compares models by balancing likelihood and complexity. Higher penalty for model complexity than AIC. Lower BIC values indicate a better fit.
	Likelihood ratio test (LRT)	Comparing the fit of two nested models. Assesses whether the more complex model provides a significantly better fit to the data.
Absolute model fit	Limited information M_2 statistic	A chi-square goodness-of-fit statistic focused on the match between the model-implied and observed frequencies of item responses.
	Root mean square error of approximation (RMSEA)	Measures the discrepancy between the model and the data per degree of freedom. Less sensitive to sample size than the M_2 statistic.
	Standardized root mean square residual (SRMSR)	Measures the average difference between the observed and model-predicted correlations of item pairs. Smaller values indicate a better fit of the model to the data.
	Posterior predictive model checking (PPMC)	Bayesian framework where data is simulated from the fitted model and compared to the observed data using various discrepancy measures.
	Transformed correlation	Examining the residual between the observed and model-predicted Fisher-transformed correlation of item pairs
Item-level fit analysis	General χ^2 goodness-of-fit test for dichotomous data	Generalized goodness-of-fit test for dichotomous data.
	Orlando and Thissen's $S - X^2$ statistic	A chi-square test statistic that compares the observed frequency of correct responses for different score groups with the frequency predicted by the model for a specific item.
	Item-level likelihood ratio G^2 test	Compare the fit of the overall model with a model where the parameters for a specific item are allowed to differ.
	Wald test	Test whether adding certain parameters to an item significantly improves the item's fit.
	General discrimination index	Validates the proposed Q-matrix by identifying how well the items q-vector discriminates between latent groups. Correctly specified q-vectors will clearly discriminate between latent groups.
	Signal detection theory informed β method	Using an AIC informed search algorithm, the β method determines how well a candidate q-vector matches the observed item responses by calculating how well the candidate q-vector discriminates between latent classes.
	Convex hull	Focus on systematic patterns in model residuals to identify Q-matrix misspecifications
	Root mean square of approximation (RMSEA)	A chi-squared based measure of item fit, computed as the square root of the weighted average, across all skill classes and item categories, of the squared differences between the estimated response probability and the observed proportion of responses, where the weights are given by the class probabilities.
	Root mean square deviation (RMSD)	Measures item fit using the square root of the average squared differences between the model-predicted response probabilities and the observed response proportions across skill classes and categories, optionally weighted across groups. Detects misfit with respect to the Item Response Function

acceptable options. In such cases, researchers must weigh multiple factors including model parsimony, theoretical considerations and practical interpretability.

This example highlights the need for more granular analysis beyond global and absolute fit statistics. When global fit indices suggest potential Q-matrix problems, researchers require diagnostic tools that can pinpoint specific items or attribute specifications that may be contributing to model misfit. This need naturally leads to item-level fit analysis, where individual items are examined for their consistency with the proposed Q-matrix structure.

2.6.2 | Item fit analysis

Item fit analysis focuses on how well each individual item is explained by the proposed CDM and Q-matrix. In general, the idea of item fit analysis is to use the results of a CDM to classify students into several groups based on their estimated latent profile and then calculate the average discrepancy between the observed and the expected responses per item. Large discrepancies suggest a misfitting item. Several statistical methods have been developed to quantify these discrepancies and identify problematic items. One foundational approach is the goodness-of-fit test proposed by Sinharay and Almond (2007), who define a general test statistic for dichotomous CDM item fit as:

$$\chi_j^2 = \sum\nolimits_{g=1}^{G} \frac{\left(O_{gj} - E_{gj}\right)^2}{E_{gj}} \sim \chi^2_{df=G-2}, \tag{21}$$

Here, G is the number of permissible latent classes, O_{gj} is the observed proportion correct and E_{gj} is the expected proportion correct. Under the assumption that the Q-matrix is correct, too many misfitting items indicate that the proposed CDM does not fit the data well. If we assume the proposed CDM is correct, the χ^2 test can be used to identify potential Q-matrix misspecification.

The chi-squared statistics in Equation 22 focus on CDMs with binary responses. Recently, Wang et al. (2024) and Quan and Wang (2024) proposed three versions of χ^2 for nominal responses. The first method, referred to a version I in Wang et al. (2024), groups students who possess the same profile g among all permissible profiles G $\left(G < 2^M\right)$, as shown in Equation 19. It is merely a polytomous extension of Equation 18. The chi-squared test statistic is computed as

$$\chi_j^2 = \sum_{g=1}^{G} \sum_{s=1}^{S_j} n_g \frac{\left(O_{jgs} - E_{jgs}\right)^2}{E_{jgk}\left(n_g - E_{jgs}\right)} \tag{22}$$

where O_{jgs} and E_{jgs} are the observed and expected count of students classified into latent group g and select option k for item j. n_g is the number of students classified into latent group g. Under the assumption that the CDM is correct, significant p-values suggest the proposed Q-vector for item j may be misspecified. Wang et al. (2024) suggested two different degrees of freedom for the reference chi-square distribution: $G \times \left(K_j - 1\right) - d_j$ and $G - d_j$ where K_j and d_j denote the number of response categories and the number of parameters for item j. Wang et al. (2024) recommended the second df as the first one can be more conservative. The second method, referred to as version II, groups students based on both the q-vector for item j and students' estimated profiles. For instance, if $q_j = (1, 1, 0, 0)$, the students with the latent classes $(1, 1, 0, 0), (1, 1, 1, 0), (1, 1, 0, 1)$ and $(1, 1, 1, 1)$ are grouped together because this item cannot differentiate them. Both versions have a limitation: for some latent groups, n_g can be very small (e.g. <5) which violates an assumption of the goodness-of-fit test.

To address this violation, Quan and Wang (2024) proposed a marginal item fit test for polytomous response, which is an extension of the Z-test in Chen et al. (2013) and De La Torre (2008). It is a marginal fit index because it no longer classifies individuals into latent groups; instead, the chi-squared test statistic is calculated as

$$\chi_j^2 = \sum_{s=1}^{S} \frac{\left(O_{js} - E_{js}\right)^2}{E_{js}}, \tag{23}$$

where O_{js} and E_{js} are the observed and expected counts for response category s within item j. Simulation studies have shown that both the version II chi-squared test in Wang et al. (2024) and the general chi-squared test in Quan and Wang (2024) are both sensitive to overfit and underfit Q-matrices when an assessment has at least 27 items. However, the general chi-squared test has a higher power to detect misfit at smaller sample sizes.

While these χ^2-based statistics provide a rigorous framework for hypothesis testing, they are sensitive to sample sizes. To address this, absolute measures of item fit, often conceptualized as effect sizes, have been used to complement hypothesis testing-based approaches. Two such indices used in CDMs are the item-level Root Mean Square Error of Approximation (RMSEA; Kunina-Habenicht et al., 2009) and the Root Mean Square Deviation (RMSD; Oliveri & von Davier, 2011). The item-level RMSEA quantifies overall discrepancy as the square root of the weighted average of squared differences between the estimated response probabilities and the observed response proportions, computed across all permissible latent skill classes and item response categories, with weights determined by the class probabilities. The RMSD, by contrast, targets misfit with respect to the item response function itself. This measures the square root of the average squared differences between model-predicted and observed response proportions across skill classes and categories. Unlike RMSEA, the RMSD can be optionally weighted across subgroups, making it particularly useful for detecting differential item functioning. Together, by capturing the absolute magnitude of discrepancy rather than relying solely on statistical significance, both indices offer researchers practical tools for evaluating item quality and guiding Q-matrix refinement (e.g. Kang et al., 2019).

In summary, CDM model misfit arises primarily from two sources: CDM model misspecification and misspecification of the Q-matrix structure. The sequential framework introduced by Chen et al. (2013) provided a systematic foundation for evaluating both CDM and Q-matrix misspecification. CDM model fit is first assessed using relative indices to determine the best-fitting CDM among candidate models. Q-matrix fit is then assessed utilizing absolute fit indices to diagnose potential misfit. This framework has since been extended through methodological innovations that enable more precise item-level diagnosis, as reviewed above. Together, these advances move beyond global evaluation toward fine-grained item diagnostics. Most recently, research has expanded into utilizing machine learning methods such as neural networks (Qin & Guo, 2023) and signal detection theory (Li & Chen, 2025). Collectively, this progression highlights a clear research trajectory from basic goodness-of-fit tests to increasingly sophisticated, data-driven approaches that provide fine-grain information on sources of misspecification.

Looking ahead, several research directions offer promising avenues for advancing CDM model fit evaluation. A critical limitation of current research is the assumption that an assessment's attribute structure (i.e. the Q-matrix) is invariant across population subgroups. Borrowing terminology from differential item functioning research, this homogeneity assumption means that when researchers propose a Q-matrix, they presume this matrix applies universally across a reference and focal group. However, without validating the proposed Q-matrix independently across these groups, it becomes impossible to distinguish between genuine Q-matrix misspecification and misfit that stems from violations of this homogeneity assumption. The challenge lies in developing validation methods that can perform effectively with the small sample sizes inherent in sub-group analyses. To address this challenge, we recommend that future research focuses on leveraging machine learning and artificial intelligence approaches, which have demonstrated promise for detecting complex patterns and relationships even when working with limited data.

A second promising avenue focuses on the response formats collected in diagnostic assessments. Current CDMs are primarily designed for binary and polytomous responses, which map relatively easily to proposed attribute structures. However, researchers are increasingly recognizing the need to extend CDMs to alternative response formats. One such example is the forced-choice (FC) format, in which respondents must select one option from two or more options rather than endorsing each item independently. By eliminating neutral response options, FC formats compel respondents to make a definitive judgement, which can reduce acquiescence bias and improve measurement precision. Recent work has established foundational frameworks for integrating FC assessments with CDMs, including the development of CDMs specifically designed for FC items (Huang, 2023) and the formalization of broader, generalized diagnostic modelling frameworks for modelling FC assessment (Nájera, Kreitchmann, et al., 2025; Nájera, Sorrel, et al., 2025).

Beyond selected-response formats, traditional CDM frameworks may also be insufficient for capturing complex cognitive processes such as logical reasoning, multi-step mathematical problem solving or written argumentation. Constructed responses of this kind are typically handled by transforming them into binary data. Recent research has begun to address this bottleneck by demonstrating the viability of ML approaches for evaluating open-ended responses; for example, Zhai et al. (2023) showed that ML algorithms can automatically extract and map natural language argumentation features onto Q-matrices.

Looking ahead, future work should continue to explore the integration of natural language processing and large language models for mapping richer constructed-response formats onto Q-matrices. Such approaches could substantially expand the diagnostic scope of CDMs, enabling more authentic assessments of complex skills while preserving the interpretability of attribute-based diagnostic inferences.

3 | DISCUSSION

In this review, we focus on six aspects of methodological advancements in CDMs that are critical to their broad applications. They are (1) accommodating polytomous attributes to allow finer gradation of attribute mastery; (2) discovering data-driven attribute relationships, which are important to discover learning sequences and facilitate personalized learning; (3) methods for faster computing, especially for a larger number of attributes; (4) methods for estimation with small sample sizes, which often occur when CDMs are applied in classroom settings; (5) Q-matrix estimation and refinement as well as (6) model fit and item fit evaluation. This review is complementary to two other latest reviews on CDM by Zhang et al. (2023) and Wang et al. (2024) as it touches upon the recent methods that aim to address the key pain points of CDM applications.

The field of CDM is evolving with more advanced models being proposed, such as the deep learning-based CDM reviewed in Wang et al. (2024). While those models still maintain the same input structure, such as students' responses, their cognitive profiles, item content and Q-matrix, it relaxes the need to pre-specify the parametric functions between item/attributes and students' responses. Instead, it uses a data-driven strategy to learn the interaction function with neural networks, thereby enjoying greater flexibility with higher predictive power to predict students' item performance. These models tend to be useful in online learning systems when on-the-fly prediction of student behaviour is essential, yet they are less useful for maintaining and replenishing the item bank because the item parameters are less interpretable. Along with the trend of infusing deep learning into psychometrics, one fruitful direction is to augment the empirical response data with item and attribute content information. For example, one can embed both contents into a common space and derive an initial content-based Q-matrix, $\boldsymbol{Q}_0$, and then use the response data to refine the Q-matrix. In addition to the Bayesian method by Culpepper (2019) that is reviewed earlier, an alternative Bayesian refinement that exploits information in $\boldsymbol{Q}_0$ can proceed by treating each $q_{jk} \in \{0, 1\}$ as a parameter with a prior informed by $\boldsymbol{Q}_0$ as follows:

$$q_{jk} \sim \text{Bernoulli}\left(\pi_{jk}\right), \text{with } logit\left(\pi_{jk}\right) = \beta_0 + \beta_1 \times 1\left\{Q_{0,jk} = 1\right\}.$$

By setting $\beta_1 > 0$, this prior preserves the content information, and one can allow item or attribute-level random effects for more flexibility, such as letting $\beta_1 \sim N_+\left(0, \sigma^2_{\beta_1}\right)$.

Another set of models that are very recently proposed and not reviewed here is deep-CDM by Gu (2024) as a form of deep generative model. Different from the deep learning-based CDM, the deep-CDM exhibits a layered architecture that implies a hierarchical generative process, with deeper layers representing more advanced and higher-order skills and shallower layers representing more basic and foundational skills. This model has the potential to model the underlying hierarchical and heterogeneous cognitive processes that generate the observed responses. The layered structure may serve as an alternative to the attribute hierarchies reviewed herein.

Also, the methodological boundary of CDMs is actively expanding to accommodate new response formats and longitudinal tracking. As discussed earlier, the adaptation of diagnostic frameworks for forced-choice and constructed-response assessments opens new areas for CDM research and application. Additionally, advancements in longitudinal CDMs, such as utilizing penalized expectation–maximization (EM) algorithms to infer learning trajectories in latent transition models (Wang, 2021), allow researchers to track attribute mastery over time dynamically.

In sum, this review underscores how recent methodological advances are expanding the scope and practicality of CDMs. By enhancing modelling flexibility, computational efficiency, small-sample applicability, fit evaluation and Q-matrix refinement, CDMs are becoming more robust tools for both large-scale testing and classroom assessment in education and in other domains. Our focus complements recent reviews by emphasizing innovations that address key barriers to application. Looking ahead, integrating CDMs with deep learning, content-informed priors and generative frameworks offers promising pathways for aligning diagnostic precision with the evolving demands of personalization such as personalized learning.

AUTHOR CONTRIBUTIONS

Chun Wang: Conceptualization; funding acquisition; writing – original draft; methodology; validation; writing – review and editing; formal analysis; project administration; supervision. **Yale Quan:** Writing – review and editing; writing – original draft; methodology. **David Arthur:** Writing – original draft; writing – review and editing; methodology.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

DISCLOSURE OF ARTIFICIAL INTELLIGENCE-GENERATED CONTENT (AIGC) TOOLS

Artificial intelligence tools (ChatGPT) were used to improve readability and grammar. No content, ideas or conclusions were generated by AI.

ORCID

Chun Wang https://orcid.org/0000-0003-2695-9781

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